CINETRACK

MOVIE, TV SHOW & ANIME TRACKER

A PROJECT REPORT

Submitted by

RISHI KUMAR R
(241401082)

in partial fulfilment of the End Semester Examination for the course

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LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY
COMPUTER SCIENCE AND BUSINESS SYSTEMS



RAJALAKSHMI ENGINEERING COLLEGE
(An Autonomous Institution, Affiliated to Anna University, Chennai)
Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

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RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled “CineTrack – Movie, TV Show & Anime Tracker” is the bonafide work of RISHI KUMAR R (241401082), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

SIGNATURE

Dr. K. R. Sowmia, Ph.D.,
HEAD OF THE DEPARTMENT
Department of Computer Science and Business Systems
Rajalakshmi Engineering College
Thandalam – 602 105

SIGNATURE

M. L. Alphin Ezhil Manuel,
COURSE FACULTY
Assistant Professor (SG)
Department of Computer Science and Business Systems
Rajalakshmi Engineering College
Thandalam – 602 105

Submitted for the End Semester Examination (Individual Project) held on _____.

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, RISHI KUMAR R (241401082), hereby declare that the project report titled "CineTrack – Movie, TV Show & Anime Tracker", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, dashboard, Deluge scripts, workflows and approval process — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature, code snippet or external resource used in this work has been duly acknowledged in the report (see Section 5.9 for the AI assistance used).

Place: Chennai
Date:

SIGNATURE OF THE STUDENT
RISHI KUMAR R
(241401082)

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RISHI KUMAR R

ABSTRACT

People who watch movies, television shows and anime usually keep track of what they have seen, what they are watching and what they plan to watch in memory, in notes on a phone or across several streaming apps. Titles are forgotten, episode progress is lost, ratings and opinions are not recorded, and there is no single view of the viewing habit.

CineTrack is a low-code application built on Zoho Creator to replace this practice. It keeps a personal media library, records each viewing session, organises titles into watchlist collections, and lets the user rate and review what they have watched. The application has three forms (Media Library, Viewing Activity and Watchlist Collections) joined by lookup fields. Reports include list reports, a Kanban Watch Progress Board with the columns Plan To Watch, Watching, Completed and Dropped, and a Viewing Statistics chart; a Dashboard page combines summary counts, a media-type chart, the Viewing Activity report and quick navigation.

Automation is written in Deluge. One script marks a title as Completed when the episodes watched reach the total episode count, another validates that the rating lies between 1 and 10, and workflows send e-mails when a title is submitted for approval and when a title is marked Completed. A Watchlist_Title_Approval process lets an approver approve or reject a newly submitted title, and the approval status is held as Pending, Approved or Rejected. A scheduled workflow, Upcoming_Watch_Reminder, runs on the Planned Watch Date of a title to remind the user. Access is controlled through a role hierarchy (CEO, Admin, Editor, User) and permission sets (Admin, CEO, Editor, User, Read, Write). A Zoho Sheet connection is configured, and the change audit trail records the actions performed on the Media_Library form.

The application was tested with sample titles including Attack On Titan, Black Clover, Breaking Bad, Stranger Things, Inception, Interstellar, Barfi, Atonement and a new submission, Dorothy, which went through the approval process. The report also records what is not complete: the reminder e-mail of the scheduled workflow is not captured in the screenshots, the Zoho Sheet connection is not yet referenced by any script, and no timing measurements or user feedback were collected, so no quantitative benefit is claimed.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Approval Process; Media Tracking

PROJECT AT A GLANCE

Attribute	Details
Project Title	CineTrack – Movie, TV Show & Anime Tracker
Application Domain	Personal media tracking (movies, TV shows, anime)
Student Name & Register No.	RISHI KUMAR R – 241401082
Zoho Creator Edition	Trial edition (application header shows “Trial expires in 2 days”)
Application Name (Link Name)	cinetrack
Published / Live URL	https://rishikumarr2024csbsra282.zohocreatorportal.in
Demo Credentials (per role)	Provided to the examiner at the demonstration (passwords are not printed in this report)
Number of Forms / Subforms	3 (Media_Library, Viewing_Activity, Watchlist Collections) / 0
Number of Reports (by type)	List: Media_Library Report, Media Watchlist, Viewing_Activity Report · Kanban: 1 (Watch Progress Board) · Pivot/Chart: Viewing Statistics, Media Types chart on the dashboard
Number of Pages / Dashboards	1 (Dashboard)
Number of Workflows	Form workflows: Auto_Complete_Title, Validate_Media_Rating, default approval status, completed-title e-mail, approval-request e-mail · Approval: 1 (Watchlist_Title_Approval) · Schedule: 1 (Upcoming_Watch_Reminder, Enabled)
Deluge Functions / Custom Actions	Deluge scripts inside form workflows; no standalone function
Integrations	Zoho Sheet connection (configured, Connected); e-mail notifications through Zoho Creator

Attribute	Details
Roles & Profiles Defined	Roles: CEO, Admin, Editor, User · Permission sets: Admin, CEO, Editor, User, Read, Write
Environments Used	Development → Stage (1.0, 01 Nov 2026) → Production (1.0, 01 Nov 2026)
Date of Production Deployment	01 November 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DS	Deluge Script (script ID used in this report)
ER	Entity–Relationship
FR / NFR	Functional / Non-Functional Requirement
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
REST	Representational State Transfer
UAT	User Acceptance Testing
UI	User Interface
UID	Unique Identifier
Zia	Zoho’s AI assistant

CHAPTER 1

INTRODUCTION

1.1 Background

Movies, television shows and anime are a common form of everyday entertainment. A regular viewer soon accumulates a long list of titles: some finished, some half-watched, many only planned. The facts that matter later are simple but easily lost: what the title is, what kind of media it is, how many episodes there are, how far the viewer has reached, when it was started and finished, how it was rated, and what the viewer thought of it.

In practice these facts are scattered across streaming-app histories, phone notes, screenshots and memory, so nobody has one place to see everything that has been watched or is planned. CineTrack is a low-code application built on Zoho Creator to provide that place. Titles are registered once in a media library, each viewing session is recorded against a title, titles can be grouped into watchlist collections, and a dashboard shows the state of the whole library. The project was built as a personal application and demonstrated with sample data.

1.2 Problem Statement

Viewers lack a single, structured and automated place to keep their movies, TV shows and anime together with the progress, rating and review of each. A simple digital system is required that stores the details of every title, tracks viewing progress and completion, keeps watchlists and collections, validates ratings, notifies the user of important changes, supports an approval step for new titles, and presents the overall position on one dashboard. The main pain points are:

- Titles are spread over several apps and notes, so the library is incomplete and hard to search.
- Episode progress is not recorded, so it is unclear where a viewer stopped.
- Ratings and reviews are not kept, or are kept without any check on their range.
- There is no automatic change of status when a title is finished, and no notification of it.
- There is no summary by media type or watch status, and no reminder for planned titles.

1.3 Objectives

- To maintain a centralised library of movies, TV shows and anime.
- To track viewing progress and episode completion.

- To maintain personal watchlists and collections.
- To rate and review watched content.
- To automate repetitive tasks using Deluge scripting.
- To send e-mail notifications and viewing reminders.
- To analyse viewing habits through reports and dashboards.

1.4 Scope of the Project

In scope. The implemented application contains three forms, list, Kanban and chart reports built from them, one Dashboard page, Deluge scripts in form workflows, e-mail notifications, an approval process for submitted titles, a scheduled reminder, a four-level role hierarchy and a set of permission sets. The application is used through a web browser.

Out of scope or not completed. The reminder e-mail sent by the scheduled workflow Upcoming_Watch_Reminder (Section 5.6) is not captured in the screenshots, so it is not claimed as tested. A Zoho Sheet connection is configured but no script uses it yet. A customer portal or mobile configuration, field-level encryption and integration with streaming services were not attempted.

1.5 Justification for a Low-Code Approach

A media tracker is a data-entry and workflow application: a few related forms, some reports, a few validations and notifications. These are exactly the building blocks that a low-code platform provides as configurable components. On Zoho Creator the forms and lookups, the list, Kanban and chart reports and the dashboard were assembled with the form builder, report builder and UI Builder, and only the status rule, the rating validation and the notifications needed Deluge. Hosting, authentication, record storage and report rendering are supplied by the platform, so effort could be spent on the tracking logic instead of infrastructure. Published studies of low-code practice report that practitioners value its ease of learning and speed of development while noting limits in customisation [1], [2].

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes with future enhancements.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The usual way of tracking viewing is informal. A viewer relies on the “continue watching” list of each streaming app, a note on the phone, a paper list, or simply memory. A title watched on one platform does not appear on another, an unfinished anime is forgotten until the viewer returns months later, and the rating given to a film is rarely written down. When the viewer wants to recall what they have seen, they scroll through several apps. The description is a general account of informal media tracking and is not a survey of a named product.

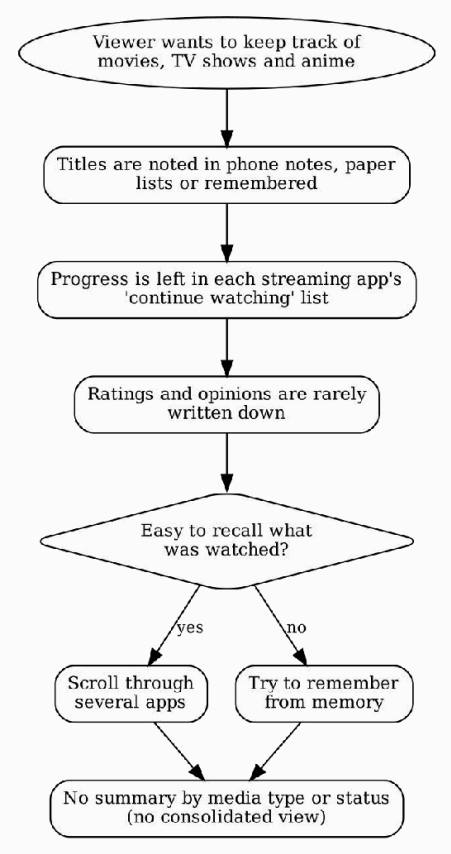


Figure 2.1: As-Is Process of Informal Media Tracking

2.2 Limitations of the Existing System

- **Scattered records.** Titles are held in several apps and notes and are easily missed or duplicated.
- **No progress tracking.** Episodes watched and the episode reached are not stored in one place.

- **No ratings or reviews.** Opinions are forgotten, and ratings are not checked against a scale.
- **No automation.** A finished title is not marked automatically and no one is notified.
- **No summary or reminders.** Counts by media type or status must be made by hand, and planned titles are not reminded.

2.3 Low-Code versus Traditional Development

Table 2.1 compares a traditional pro-code implementation of the tracker with the low-code implementation on Zoho Creator. The comparison is qualitative. No development-time or cost measurements were recorded, so none are claimed.

Table 2.1: Comparison of Low-Code and Traditional Development for CineTrack

Criterion	Traditional (Pro-Code)	Low-Code on Zoho Creator
Development time	Database design, back-end services, user interface and deployment must each be built and integrated.	Forms, reports and the dashboard are assembled in builders; only short Deluge scripts were written.
Skills required	A front-end and a back-end language, SQL, hosting and security administration.	Familiarity with the platform and basic Deluge scripting.
Hosting	Servers or cloud services must be provisioned and maintained.	Hosted and maintained by the platform.
Mobile	A separate mobile client or responsive front end.	The builder offers desktop, tablet and phone previews, and the application was viewed in the phone layout (Figure 5.19); a dedicated mobile app configuration was not done.
Security and access	Authentication, sessions and role logic written in code.	Roles and permission sets configured in Settings.
Change and maintenance	Code change, testing and redeployment.	Fields, reports and workflows edited in the builders.
Customisation limits	Practically unlimited.	Bounded by platform components and Deluge (Section 8.4).
Cost	Developer time plus infrastructure and licences.	Platform subscription; built on a trial edition (header shows “Trial expires in 2 days”).

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Source & Year	Title / Product	Key Findings	Relevance
1	Y. Luo, P. Liang, C. Wang, M. Shahin and J. Zhan, 2021 [1]	Characteristics and Challenges of Low-Code Development: The Practitioners’ Perspective	Empirical study of practitioners’ discussions of low-code advantages and challenges.	Supports the choice of low-code and Section 8.4.

S.No.	Source & Year	Title / Product	Key Findings	Relevance
2	M. A. Al Alamin et al., 2021 [2]	An Empirical Study of Developer Discussions on Low-Code Software Development Challenges	Empirical study of developer questions about low-code platforms, grouped into topics.	Background for limitations in Sections 8.4 and 8.5.
3	Zoho Corporation [3]	Zoho Creator Help	Documentation of forms, reports, pages, workflows, approvals, connections and permissions.	Reference for Chapters 4 and 5.
4	Zoho Corporation [4]	Deluge Script reference	Documentation of the Deluge language.	Reference for Section 5.5 and Appendix C.

2.5 Emerging Trends in Low-Code Development

Three general trends are relevant. First, low-code platforms increasingly embed AI assistance in their builders; the Zia assistant icon is visible on the Zoho Creator builder screens, although no Zia feature is part of CineTrack. Second, platforms emphasise integration with the surrounding service ecosystem, which this project touches through a Zoho Sheet connection. Third, as more business users build applications, governance concerns such as role design and audit trails become more important; this project applies them through permission sets, a role hierarchy and the change audit trail. These trends are stated qualitatively and are not backed by market figures.

2.6 Proposed System

The proposed system is CineTrack. Titles are registered once in the Media Library with their type, genre, episodes, status, rating and review. Each viewing session is stored in Viewing Activity against a title, and titles are grouped in Watchlist Collections. A Deluge script marks a title Completed when all episodes are watched, a second script keeps ratings within 1 to 10, an e-mail is sent when a title is completed, and a newly submitted title goes through an approval process. Reports, a Kanban board and a dashboard summarise the library. This removes the scattered notes, records progress, ratings and reviews, automates completion, and gives one view of the whole library.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities
CEO (role)	Top level of the role hierarchy; permission set CEO gives access to everything.	Oversight of all records and settings.
Admin	Owner of the application; approves or rejects submitted titles.	Manage users, roles and permissions; view all media records, reports, dashboards and viewing activity.
Editor	Reports to Admin in the hierarchy.	Add movies, TV shows and anime, edit media details, update watch progress and add viewing activity.
User	Viewer of the library.	View available media titles and reports; view own watchlist and viewing history.

3.2 Functional Requirements

FR-01 to FR-11 correspond to features built in the application; for FR-11 the schedule is enabled but its reminder e-mail is not yet captured. The IDs are used again in the test cases of Chapter 6.

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall store each title with Title Name, Media Type and Genre mandatory, and Release Year, Director / Studio, episode counts, rating, dates and review optional.	Editor, Admin	High
FR-02	The system shall record each viewing session (title, activity date, episodes watched today, episode reached, notes) against a title.	Editor, Admin	High
FR-03	The system shall group titles into watchlist collections with category, priority and status.	Editor, Admin	Medium
FR-04	The system shall set Approval Status to Pending when a new title is added.	System	High
FR-05	The system shall set Watch Status to Completed when Episodes Watched is not less than Total Episodes.	System	High
FR-06	The system shall reject a My Rating outside 1 to 10 and cancel the submission.	System	High
FR-07	The system shall send an e-mail when a title is marked Completed.	System	Medium
FR-08	The system shall send an e-mail to the approver when a new title is submitted, and allow Approve or Reject.	System, Admin	High
FR-09	The system shall provide a list report, a Kanban board by Watch Status and a chart by media type, and a dashboard that combines summaries.	All roles	Medium

FR ID	Requirement	Role(s)	Priority
FR-10	The system shall restrict users through a role hierarchy and permission sets.	Admin	Medium
FR-11	The system shall send a scheduled reminder for titles whose Planned Watch Date has come, if approved and not yet watched.	System	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and the dashboard shall open without perceptible delay for the small data volume of the demonstration. No measurement was recorded.
NFR-02	Security	Users shall reach only the forms, reports and pages allowed by their permission set.
NFR-03	Usability	A title shall be added from one form using drop-downs, date pickers and a review box; the application shall run in a standard web browser.
NFR-04	Availability	The application relies on the availability of the Zoho Creator service; no separate hosting is maintained.
NFR-05	Maintainability	Business rules shall be held in named form workflows so that they can be changed without redesigning the forms.
NFR-06	Auditability	Record creation and edits shall be traceable through the platform's audit trail (Section 7.4).

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator, trial edition (India data centre, creator.zoho.in)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome (latest version)
Other Zoho services	Zoho Sheet (connection configured); e-mail notifications sent through Zoho Creator
Mobile	Not configured; the builder offers desktop, tablet and phone preview toggles
Client hardware	Any computer able to run a current web browser, with an internet connection

3.5 Use Case Model

The Editor adds titles, records viewing activity, creates collections and updates watch progress. The Admin also manages roles and permissions and reviews titles submitted for approval. The User views titles, the watchlist and the viewing history. The e-mail notification and the approval request are sent by the system, and the Zoho Sheet connection

is shown as a secondary service.

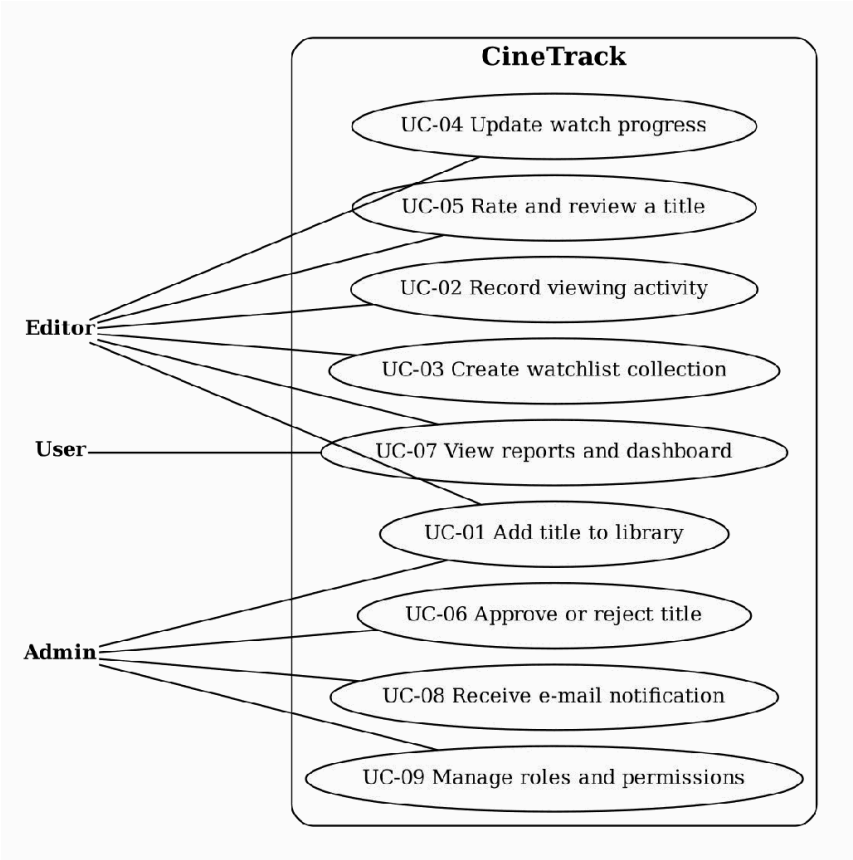


Figure 3.1: Use Case Diagram of CineTrack

Table 3.5: Use Case Description – Add Title (UC-03)

Item	Description
Use Case ID / Name	UC-03 – Add Title to Media Library
Actor(s)	Editor (primary); Admin
Pre-condition	The actor has access to the Media_Library form.
Main Flow	1. The actor opens Media_Library → Media_Library. 2. Enters Title Name, Media Type, Genre and optional details. 3. Clicks Submit. 4. The system sets Approval Status to Pending, applies the completion and rating rules, saves the record and e- mails the approver.
Alternate Flow	If Title Name, Media Type or Genre is empty, or My Rating is outside 1 to 10, the form does not submit and shows a message.
Post-condition	The record appears in Media_Library Report and Media Watchlist, and a task appears in Approvals → Pending Tasks.
Related FR	FR-01, FR-04, FR-05, FR-06, FR-08

Table 3.6: Use Case Description – Approve or Reject (UC-04)

Item	Description
Use Case ID / Name	UC-04 – Approve or Reject a Submitted Title
Actor(s)	Admin (approver)
Pre-condition	A title has been submitted and an approval task exists.

Item	Description
Main Flow	1. The approver opens the e-mail or Approvals → Pending Tasks. 2. Opens the Approval Summary. 3. Clicks Approve or Reject. 4. The task moves to Completed Tasks.
Post-condition	The task is shown under Approvals → Completed Tasks.
Related FR	FR-08

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The application runs entirely inside Zoho Creator; it has no custom front end, separate back end or external database. Users reach it through a web browser. The presentation layer consists of the three forms, the reports and the Dashboard page. The logic layer consists of form workflows holding Deluge scripts: the validation scripts run when a form is validated on submission (Auto_Complete_Title and Validate_Media_Rating), and the notification actions run after a successful submission. The data layer is the set of Media Library, Viewing Activity and Watchlist Collections records, related through lookup fields. A role hierarchy and permission sets govern access, an approval process governs new titles, and a Zoho Sheet connection is available for integration.

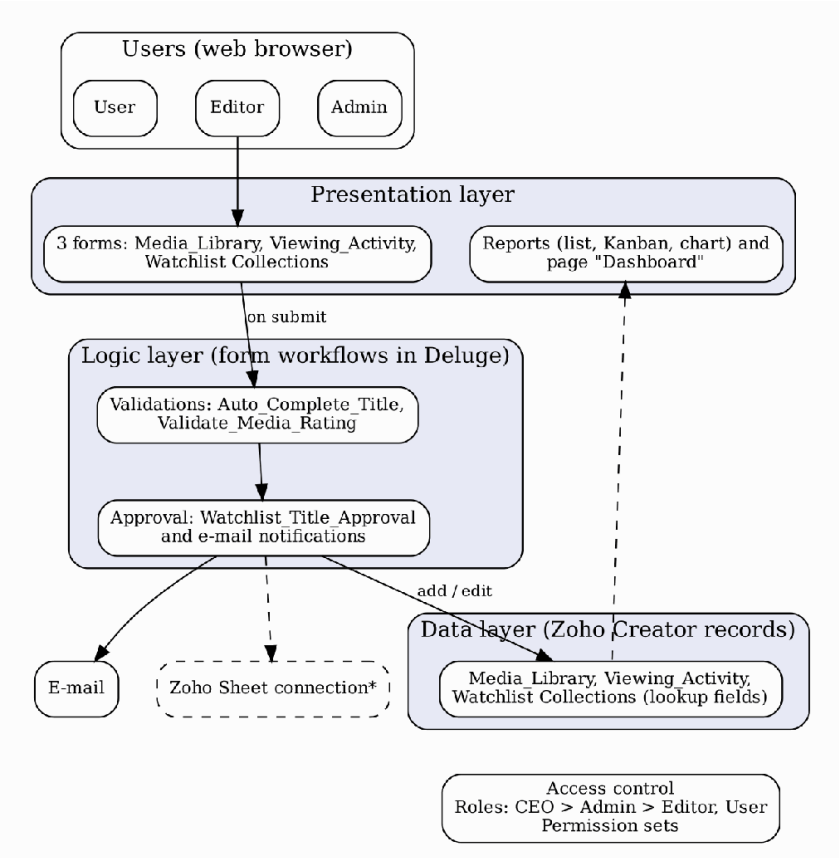


Figure 4.1: System Architecture of CineTrack

4.2 Data Model

Zoho Creator stores data as records of forms. A form plays the role of a table, a field plays the role of a column, and a lookup field plays the role of a foreign key. Media Library holds

each title. Viewing Activity carries a Title lookup, so one title can have many viewing sessions. Watchlist Collections also carries a Title lookup, so a title can appear in collections. No subform is used because each one-to-many relationship is between records that are useful on their own.

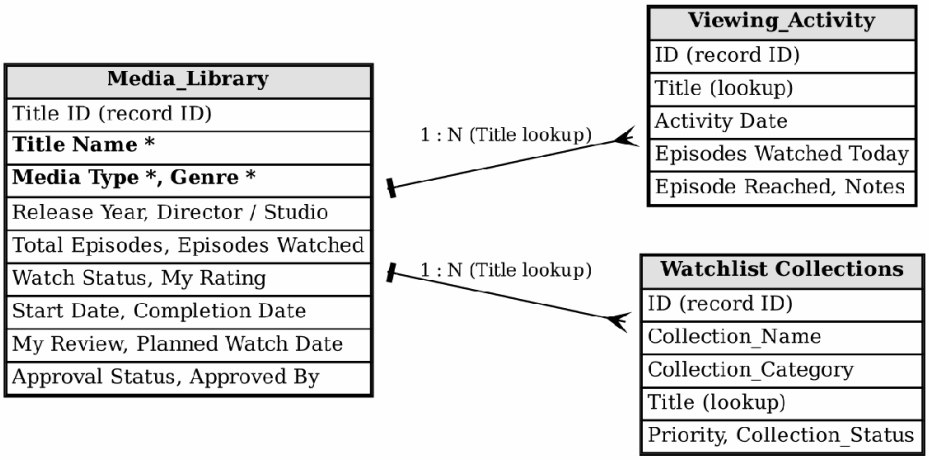


Figure 4.2: Entity–Relationship Diagram of CineTrack

Data consistency is supported in four ways: titles are chosen through lookups so an activity cannot refer to a title that does not exist; Media Type, Genre, Watch Status, Approval Status and the collection fields are drop-downs with fixed values; Title Name, Media Type and Genre are mandatory; and the rating is validated by a script.

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship
Media_Library	Title Name, Media Type, Genre, Release Year, Watch Status, My Rating, Approval Status	Viewing_Activity; Watchlist Collections	One title, many activities and collections (Title lookup)
Viewing_Activity	Title, Activity Date, Episodes Watched Today, Episode Reached, Notes	Media_Library	Many-to-One
Watchlist Collections	Collection_Name, Collection_Category, Title, Priority, Collection_Status	Media_Library	Many-to-One

4.3 Form Design

Field labels, types and mandatory flags below are taken from the live-form screenshots in Chapter 5. Link names follow Creator’s default convention (spaces replaced by underscores) and should be confirmed in the builder where they are not visible in a screenshot.

Table 4.2: Field Specification – Media_Library Form

Field Label	Field Type	Mandatory	Validation / Remarks
Title Name	Single Line	Yes	—
Media Type	Drop Down	Yes	Movie, Anime, TV Show
Genre	Drop Down	Yes	e.g. Action, Drama, Horror, Sci-Fi, Thriller
Release Year	Number	No	Number format #######
Director / Studio	Single Line	No	—
Total Episodes	Number	No	Used by Auto_Complete_Title
Episodes Watched	Number	No	Used by Auto_Complete_Title
Watch Status	Drop Down	No	Plan To Watch, Watching, Completed, Dropped
My Rating	Number	No	Must be 1 to 10 (Validate_Media_Rating)
Start Date / Completion Date	Date	No	Format dd-MMM-yyyy
My Review	Multi Line	No	Personal review
Planned Watch Date	Date	No	Used by the planned reminder
Approval Status	Drop Down	No	Pending, Approved, Rejected
Approved By	Single Line	No	Approver name

Table 4.3: Field Specification – Viewing_Activity Form

Field Label	Field Type	Mandatory	Remarks
Title	Lookup (Media_Library)	No	Selected title
Activity Date	Date	No	dd-MMM-yyyy
Episodes Watched Today	Number	No	—
Episode Reached	Number	No	—
Notes	Multi Line	No	Free-text note

Table 4.4: Field Specification – Watchlist Collections Form

Field Label	Field Type	Mandatory	Remarks
Collection_Name	Single Line	No	Name of the collection
Collection_Category	Drop Down	No	Category of the collection
Title	Lookup (Media_Library)	No	Selected title
Priority	Drop Down	No	Priority of the title
Collection_Status	Drop Down	No	Status of the collection

The planned design also lists Date Added and Notes on the Watchlist Collections form; they are not visible in the form screenshot and should be added or removed from this table to match the application.

4.4 Process and Workflow Design

Title submission. Before the record is saved, the validation workflows run: Validate_Media_Rating cancels the submission if the rating is outside 1 to 10, and Auto_Complete_Title sets Watch Status to Completed if all episodes are watched. The approval status is set to Pending, the record is saved, and after saving an e-mail is sent to the approver and an approval task is created. The approver opens the Approval Summary and chooses Approve or Reject. When a title is marked Completed, a notification e-mail is sent.

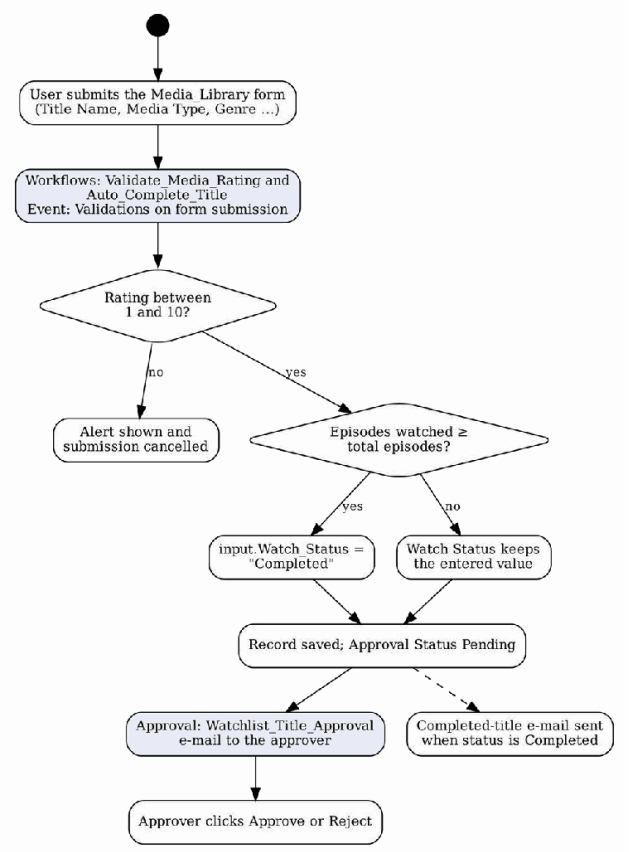


Figure 4.3: Workflow / Process Flow for Submitting a Title

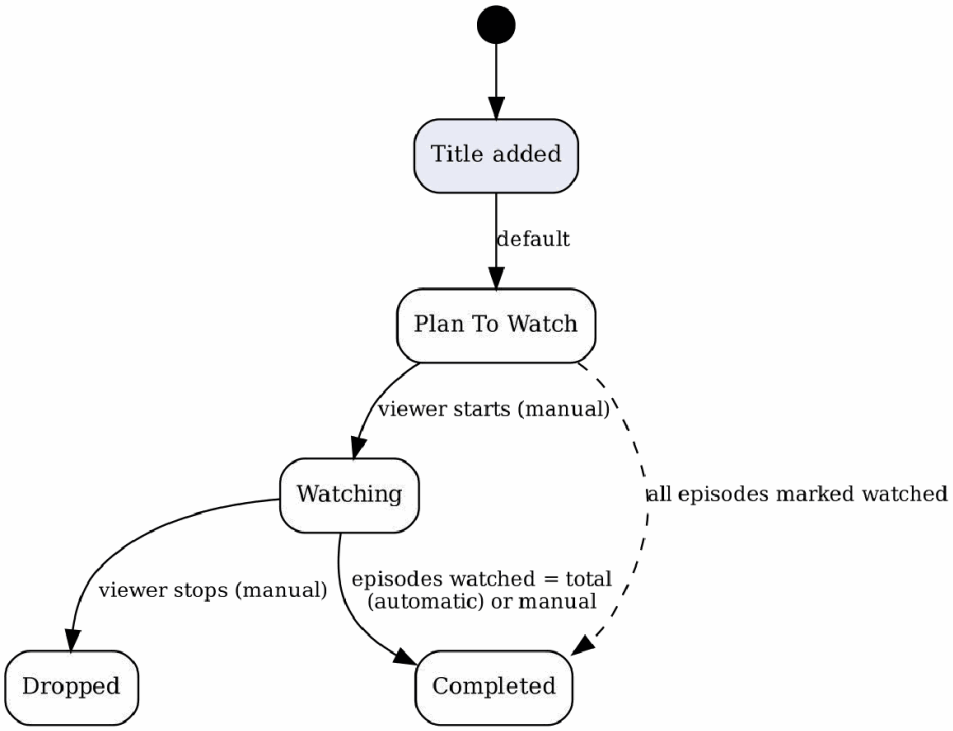


Figure 4.4: Status Lifecycle of a Title

Status lifecycle (Figure 4.4). A title moves through the Watch Status values Plan To Watch, Watching and Completed, or Dropped. Completed is also set automatically by script. The order is a drop-down choice and is not otherwise enforced.

Table 4.5: Watch Status Values and Transitions

Status	Entered When	Changed By
Plan To Watch	The title is added and not yet started.	User / default
Watching	The viewer has started the title.	Editor (manual)
Completed	Episodes watched reach the total, or the viewer marks it.	Auto_Complete_Title (automatic) or Editor
Dropped	The viewer stops watching.	Editor (manual)

4.5 User Interface and Navigation Design

The menu has the sections Media_Library (Media_Library form, Media_Library Report, Media Watchlist, Viewing Statistics, Watch Progress Board), Approvals (Pending Tasks, Completed Tasks), Viewing_Activity, the Dashboard page, and Collections. The interface uses the platform’s dark side menu and standard form layout.

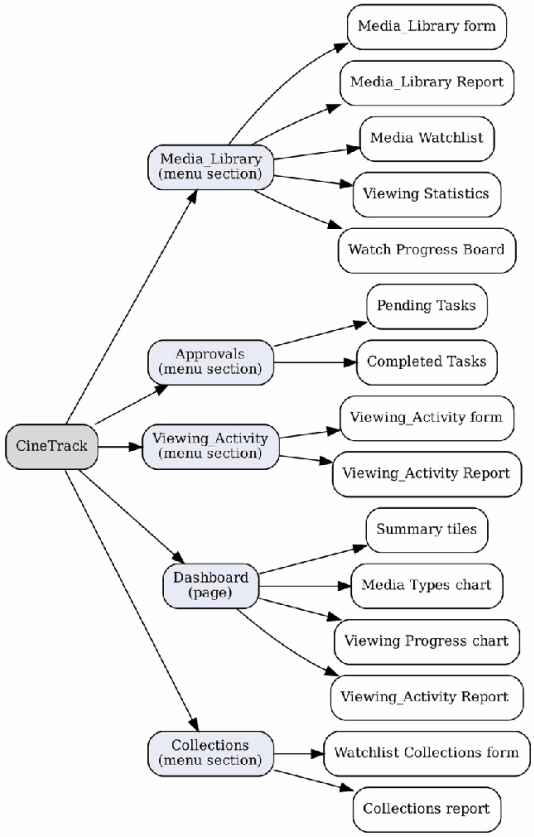


Figure 4.5: Navigation Structure of the Application

4.6 Access Control Design

Access is designed on three levels: Zoho sign-in, a role hierarchy (CEO, then Admin, then Editor and User), and permission sets that define what each kind of user may do. The Admin permission set allows viewing, adding, editing and deleting all media records, approving or rejecting titles, managing users, roles and permissions, and viewing all reports, dashboards and viewing activity. The Editor set allows adding titles, editing media details, updating progress and adding viewing activity. The User set allows viewing titles and reports and the user’s own watchlist and history. Read and Write are built-in sets, and CEO allows access to everything.

CHAPTER 5

IMPLEMENTATION

5.1 Application Setup

The application was created from scratch in a Zoho Creator account (India data centre) under the name CineTrack. Three forms were created, and the menu was organised into the sections Media_Library, Approvals, Viewing_Activity, Dashboard and Collections. The application header shows that the account was on a trial edition when the screenshots were taken. The application address is https://rishikumarr2024csbsra282.zohocreatorportal.in.

5.2 Forms and Data Capture

Media_Library form. This is the main form. Title Name, Media Type and Genre are mandatory. It also holds Release Year, Director / Studio, Total Episodes, Episodes Watched, Watch Status, My Rating, Start Date, Completion Date, My Review, Planned Watch Date, Approval Status and Approved By (Figure 5.1).

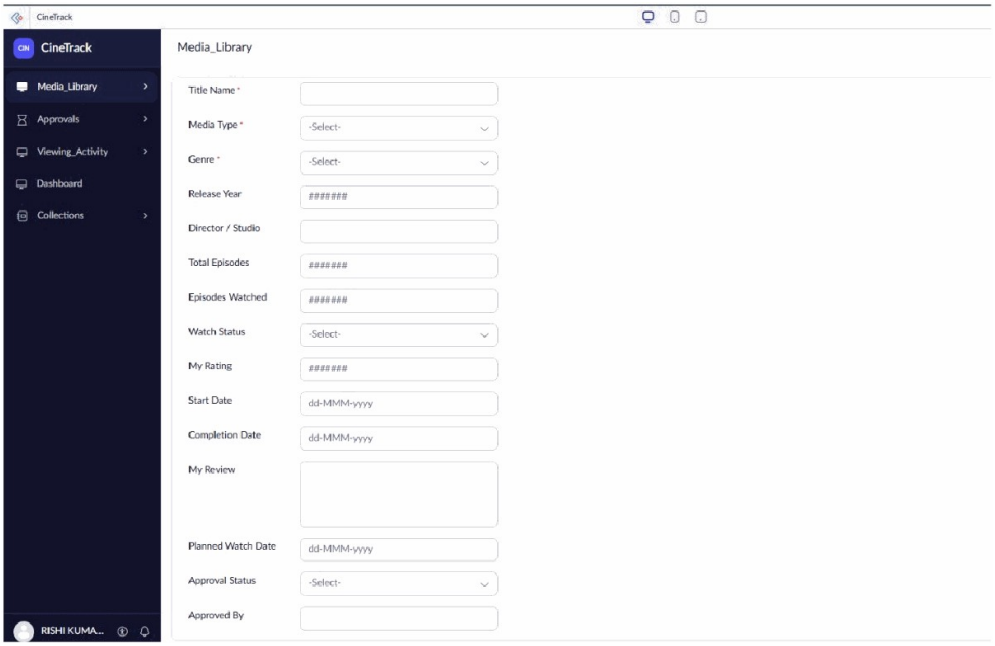


Figure 5.1: Media_Library Form in Live Mode

Viewing_Activity form. It records each viewing session with a Title lookup, Activity Date, Episodes Watched Today, Episode Reached and Notes (Figure 5.2).

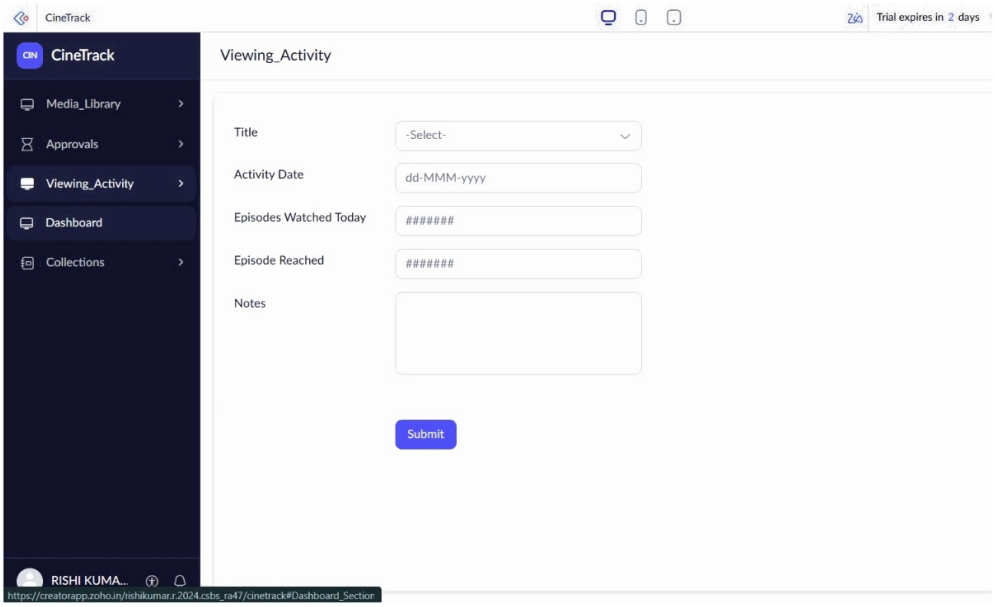


Figure 5.2: Viewing_Activity Form in Live Mode

Watchlist Collections form. It organises titles into collections with Collection_Name, Collection_Category, Title (lookup), Priority and Collection_Status. Its submit button is labelled Watchlist (Figure 5.3).

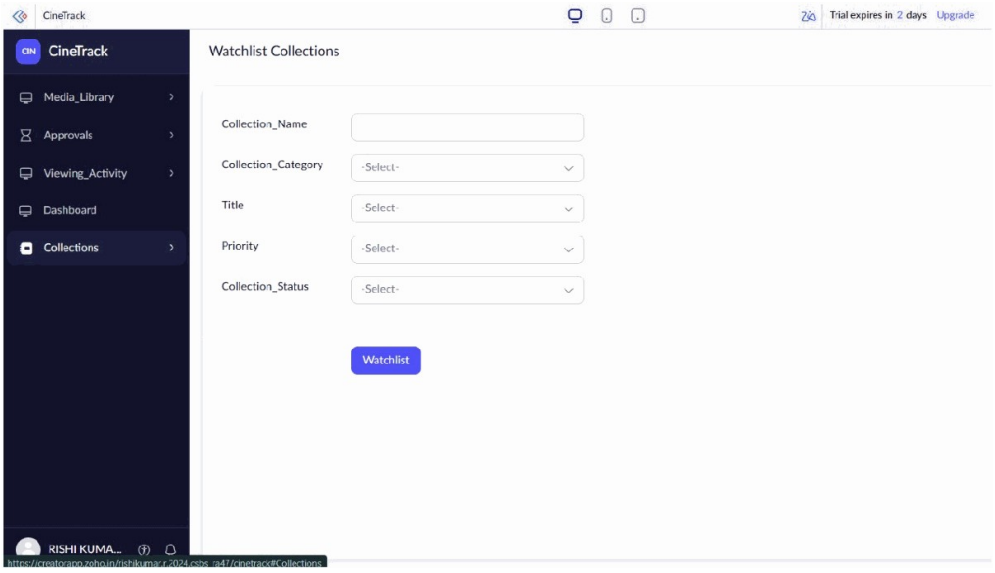


Figure 5.3: Watchlist Collections Form in Live Mode

5.3 Reports

Creator generated a list report for each form. Three kinds of report were used for analysis.

- **Media_Library Report and Media Watchlist (list reports).** Media_Library Report shows Title ID, Title Name, Media Type, Genre, Release Year, Director / Studio and Total Episodes (Figure 5.4). Media Watchlist shows Title Name, Media Type, Watch Status and My Rating for all registered movies, TV shows and anime

(Figure 5.5).

- **Viewing Statistics / Media Type analysis (chart).** A chart report compares the number of movies, TV shows and anime in the library; the same breakdown appears as a donut chart on the Dashboard (Figure 5.7).
- **Watch Progress Board (Kanban report).** Titles are shown as cards in the columns Plan To Watch, Watching, Completed and Dropped (Figure 5.6). In the screenshot the Dropped column has no records.

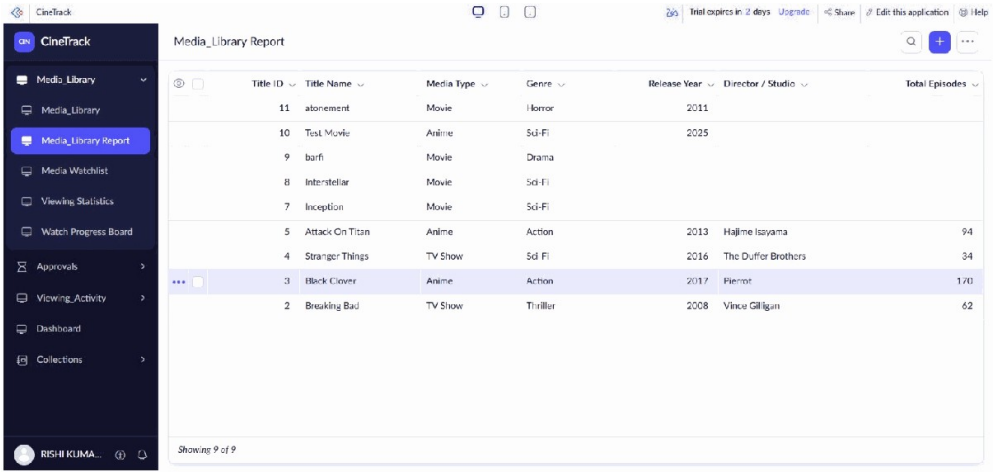


Figure 5.4: Media_Library Report (9 of 9 records)

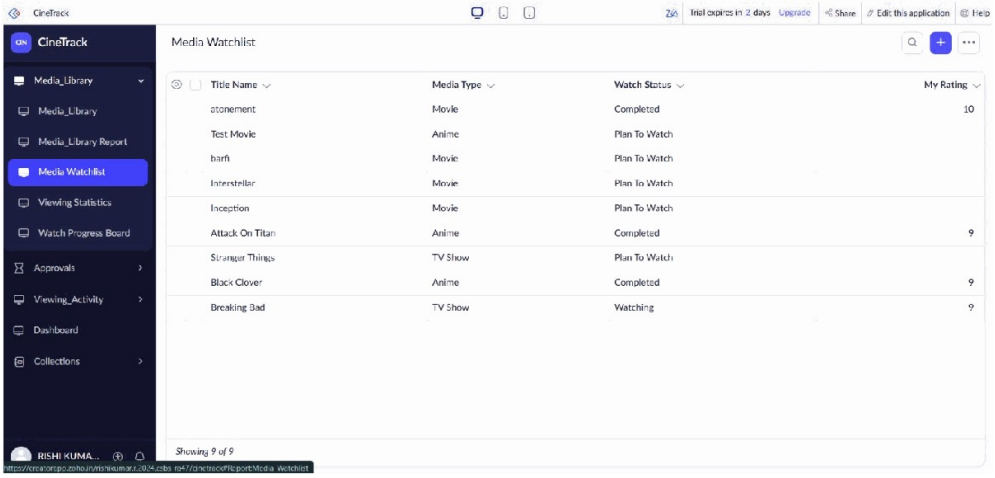


Figure 5.5: Media Watchlist Report (9 of 9 records)

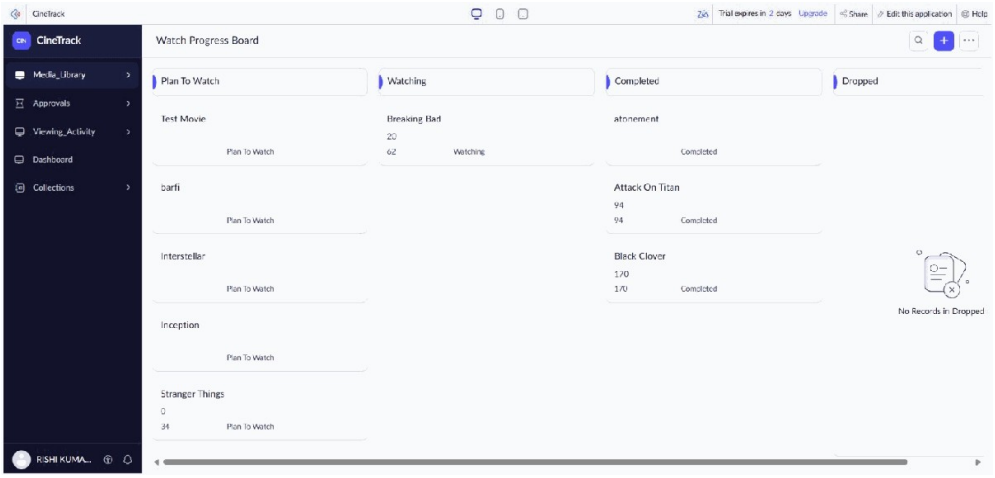


Figure 5.6: Watch Progress Board (Kanban Report)

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Purpose
Media_Library Report	List	Media_Library	All titles with type, genre, year, studio and episodes
Media Watchlist	List	Media_Library	Title, type, watch status and rating
Watch Progress Board	Kanban	Media_Library	Cards grouped by Watch Status
Viewing Statistics	Chart	Media_Library	Count of titles by media type
Viewing_Activity Report	List	Viewing_Activity	Viewing history (embedded on the Dashboard)

5.4 Pages and Dashboards

The Dashboard page was built in the UI Builder. It shows summary tiles (Total Titles, Completed, Currently Watching, Plan to Watch and Average Rating), a Media Types donut chart, an embedded Viewing_Activity Report, a Viewing Progress bar chart of titles by watch status, a Latest Updates panel and a Navigate panel with quick actions (My Library, Add Title, Viewing Activity, Watchlist). In the screenshot the tiles read 10 total titles, 3 completed, 1 watching, 6 planned and an average rating of 9 out of 10 (Figure 5.7). A later phone view of the Dashboard shows 11 titles and 4 completed, after more titles were added (Figure 5.19).

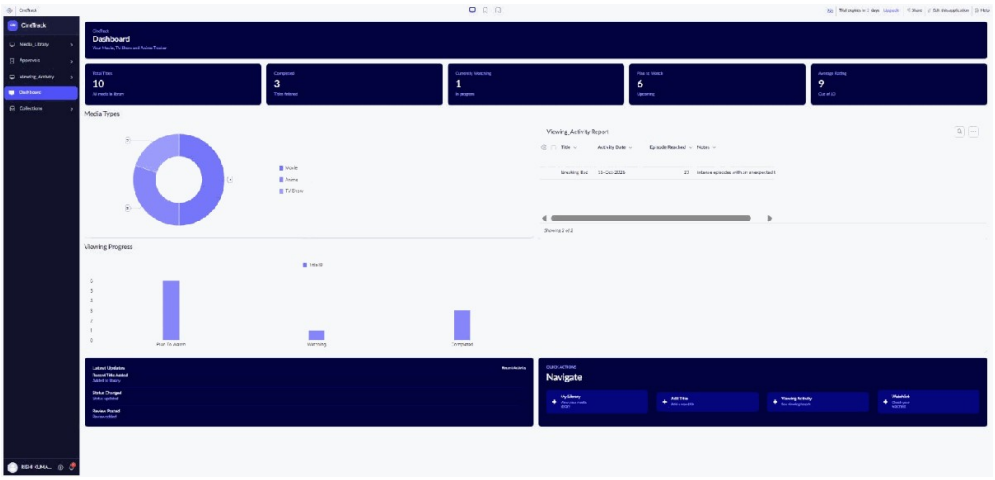


Figure 5.7: CineTrack Dashboard

5.5 Deluge Scripting

Deluge scripts are configured as actions inside form workflows on the Media_Library form. Table 5.2 lists them; the scripts visible in the workflow editor are given in Appendix C.

Table 5.2: Deluge Scripts and Workflows Implemented

Script / Workflow	Trigger	Purpose	Deluge Concepts
Auto_Complete_Title	Media_Library → Created or Edited → Validations on form submission	Set Watch Status to Completed when Episodes Watched ≥ Total Episodes	if, null checks, comparison, assignment to input.Watch_Status
Validate_Media_Ratin g	Media_Library → Created or Edited → Validations on form submission	Reject a rating outside 1 to 10	if, logical OR, alert, cancel submit
Set default approval status	Media_Library → Created	Set Approval Status to Pending for a new title	assignment to input field
Completed title notification	Media_Library → status becomes Completed	E-mail the user that a title is Completed	sendmail
Watchlist approval request	Media_Library → Created (approval)	E-mail the approver about a new title	sendmail
Upcoming_Watch_Re minder	Schedule on Media_Library, based on Planned Watch Date (1 action)	Remind the user about a title planned to be watched	Schedule workflow

Auto_Complete_Title first checks that Total Episodes has a value, then that it is greater than 0 and Episodes Watched has a value, and finally sets Watch Status to Completed if the episodes watched are at least the total (Figure 5.8). Validate_Media_Rating checks that My Rating is not null and shows the alert “Rating must be between 1 and 10.” and cancels the submission if it is below 1 or above 10 (Figure 5.9).

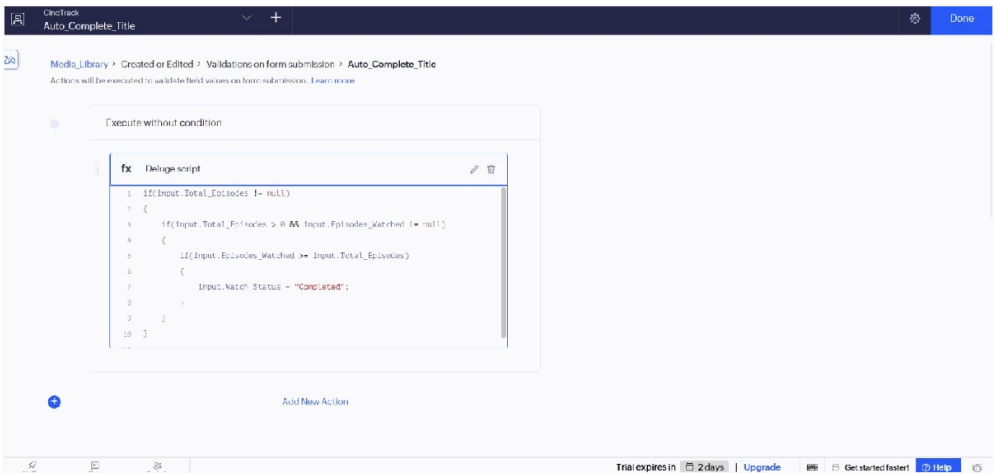


Figure 5.8: Deluge Script Auto_Complete_Title

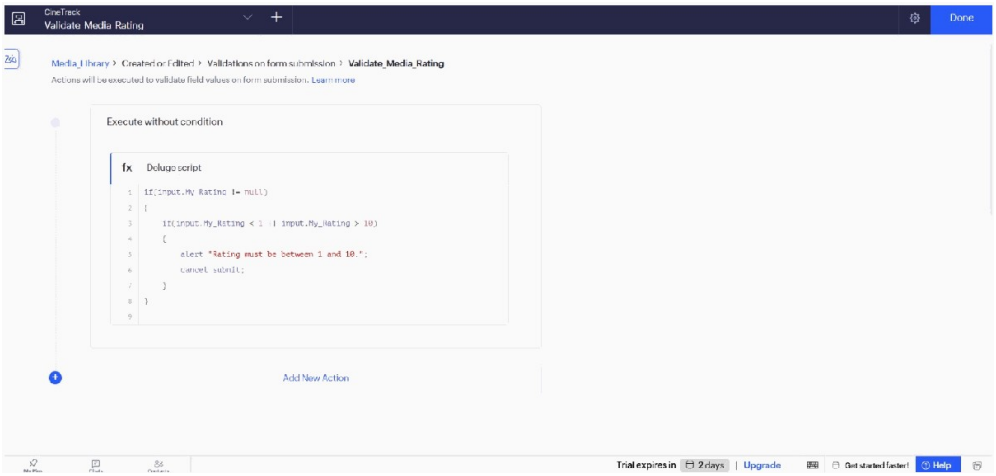


Figure 5.9: Deluge Script Validate_Media_Rating

The e-mails received are described in Section 8.1. Standalone functions, fetch statements and loops were not needed so far and are not used.

5.6 Workflows and Automation and Approvals

Form workflows. The five automations in Table 5.2 are form workflows on Media_Library. Validation workflows run before saving so that a value can be changed or the submission cancelled, and notification workflows run after saving.

Approval process. When a new title is submitted, the approval process Watchlist_Title_Approval creates a task for the approver, who opens an Approval Summary showing Title ID, Title Name, Media Type, Genre, Release Year, Director / Studio and Total Episodes, and clicks Approve or Reject (Figure 5.12). The task is listed under Approvals → Pending Tasks (Figure 5.10) and, once reviewed, under Completed Tasks (Figure 5.11). The approval status is kept as Pending, Approved or Rejected.

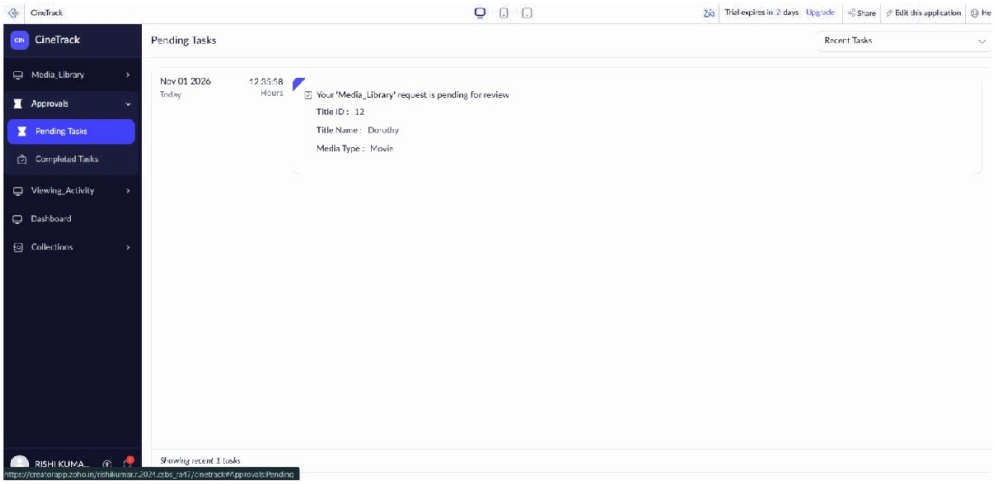


Figure 5.10: Approvals – Pending Tasks (Title ID 12, Dorothy)

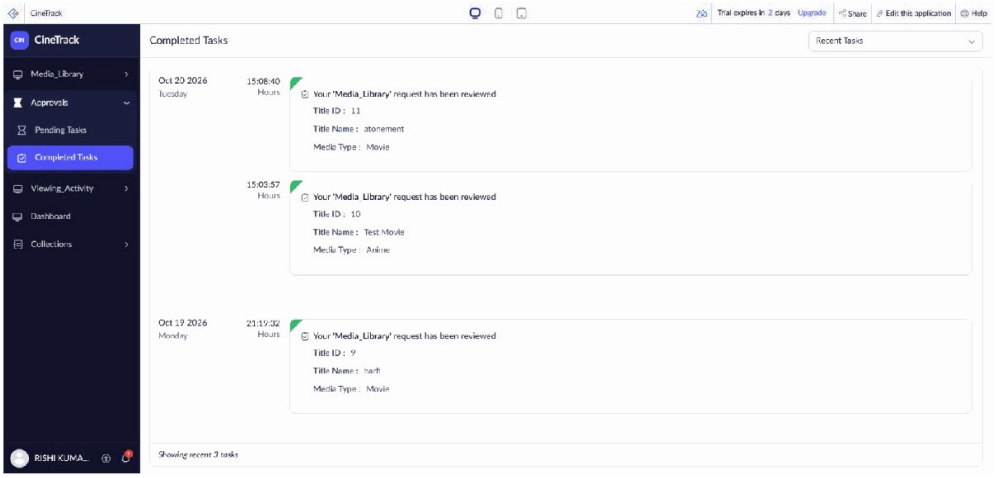


Figure 5.11: Approvals – Completed Tasks (Title IDs 9, 10 and 11)

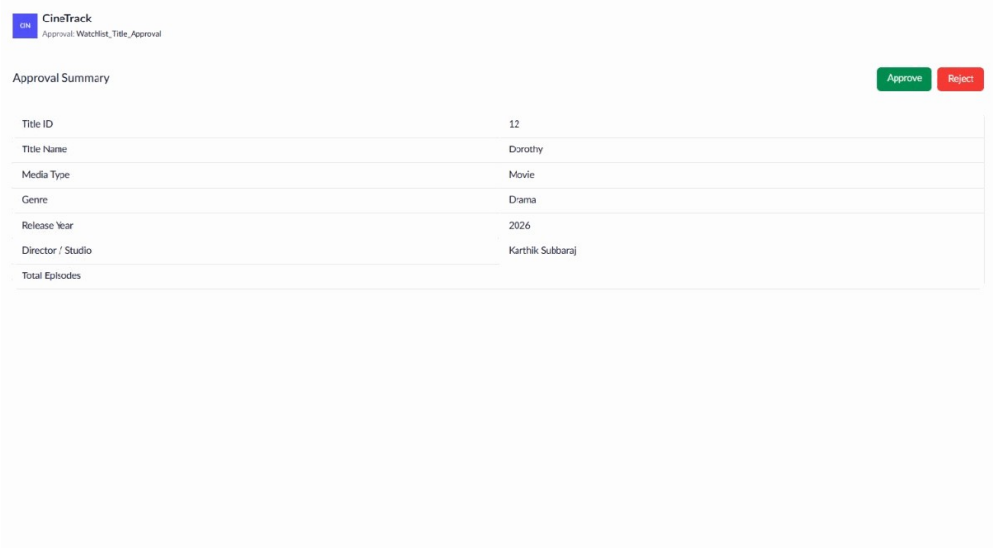


Figure 5.12: Approval Summary with Approve and Reject Buttons

The workflow lists show the approval Watchlist_Title_Approval on Media_Library, triggered On review, with 4 actions and the status Enabled (Figure 5.13).

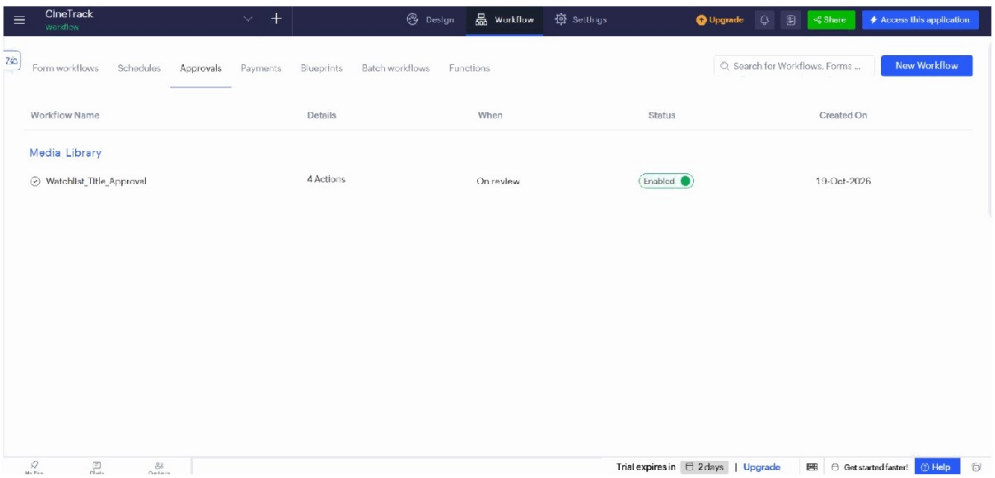


Figure 5.13: Approvals Tab – Watchlist_Title_Approval (4 Actions, Enabled)

Scheduled reminder. The schedule Upcoming_Watch_Reminder on Media_Library runs on the Planned Watch Date, has one action, is Enabled and was created on 19-Oct-2026 (Figure 5.14). It is designed to check the Planned Watch Date, Approval Status and Watch Status of a title and to e-mail the user about approved titles that are still Plan To Watch. The reminder e-mail itself is not captured in the screenshots, so it is not claimed as tested.

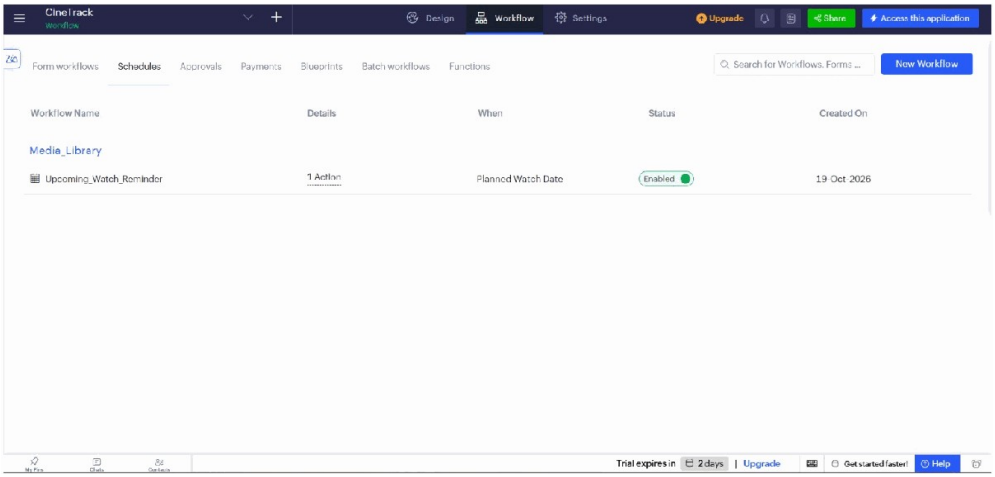


Figure 5.14: Schedules Tab – Upcoming_Watch_Reminder (Enabled)

5.7 Integrations

A connection named Zoho Sheet Connection with the link name zoho_sheet_connection exists under Microservices → Connections and shows the status Connected (Figure 5.15). Its permissions include All Data Operations and Create Record, the authorization is the same for all environments, and the owner is the Administrator. Connection access is switched on for CineTrack, and the page shows “No References”, which means no script uses the connection yet (Figure 5.16). E-mail notifications are sent by the platform from notifications@trial.zohocreatormail.com. No REST call with invokeurl, webhook or Creator REST API was used.

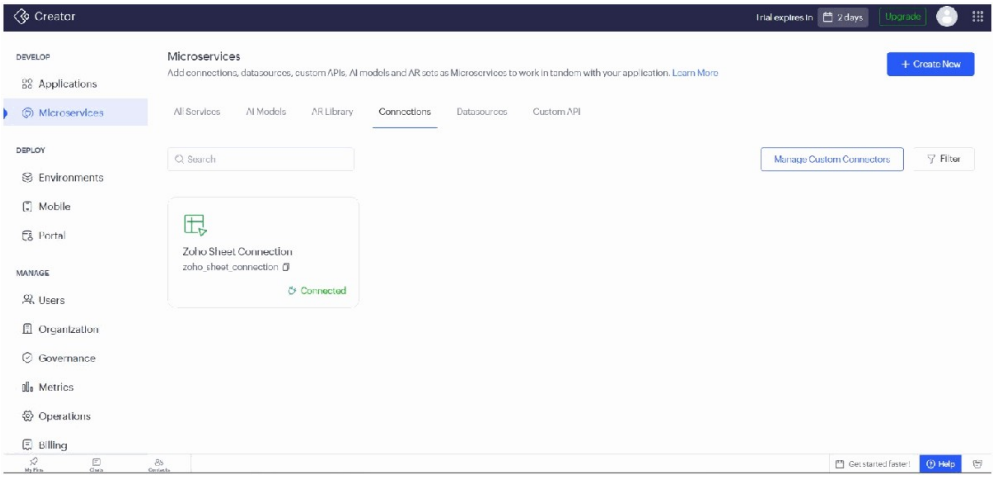


Figure 5.15: Connections List Showing Zoho Sheet Connection (Connected)

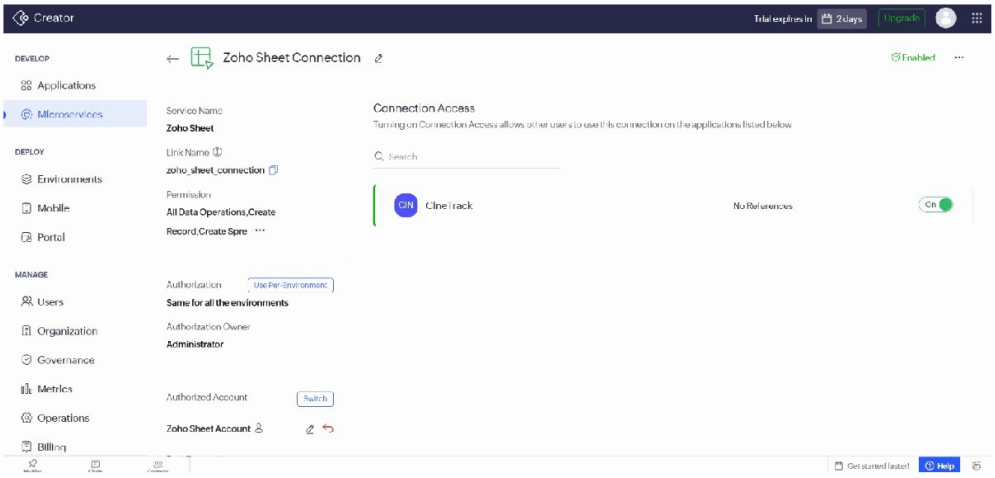


Figure 5.16: Zoho Sheet Connection Details and Access Turned On for CineTrack

Table 5.3: Integrations

Integration	Service	Method	Status
Spreadsheet link	Zoho Sheet	Creator connection zoho_sheet_connection	Connected; no references in the application yet
E-mail notification	Zoho Creator mail	Workflow e-mail actions	Working; e-mails received (Section 8.1)

5.8 User Management and Role-Based Access

Under Settings → Users, the Permissions tab lists the permission sets Admin, CEO, Editor, Read, User and Write (Figure 5.17). The Roles tab holds the hierarchy CEO → Admin → Editor and User (Figure 5.18).

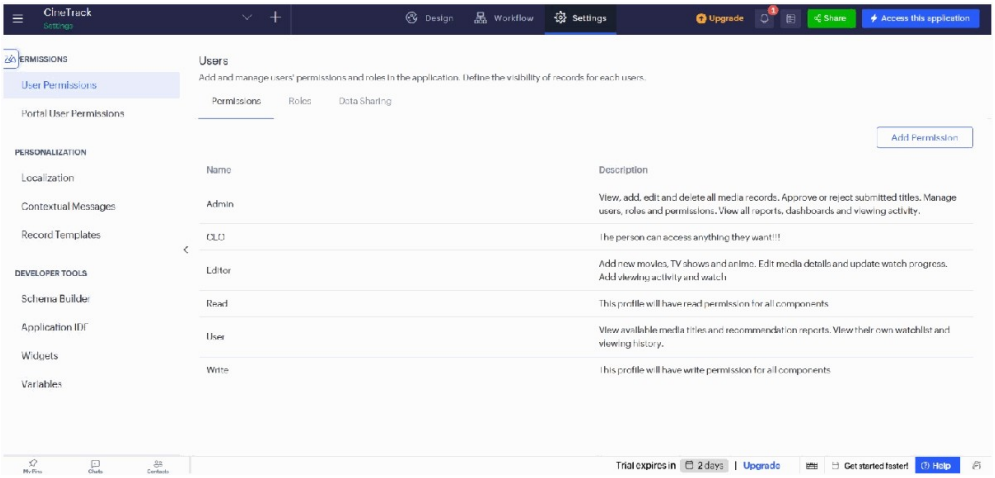


Figure 5.17: Permission Sets of CineTrack

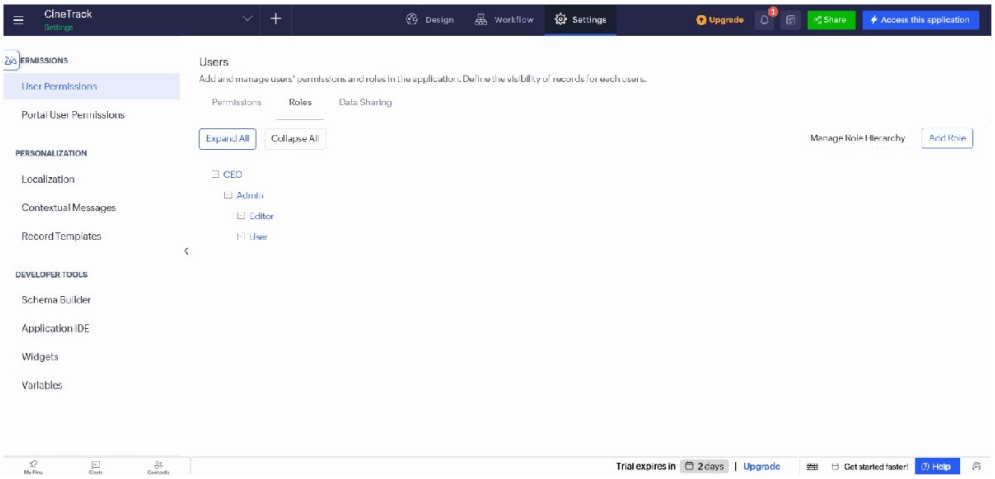


Figure 5.18: Role Hierarchy: CEO → Admin → Editor, User

What is not done. Whether users were invited and assigned to each role, and whether the application was tested by signing in as each role, is not evidenced by the screenshots; record this at the demonstration. No record-level criteria such as “Added User == zoho.loginuser” are shown, so the User set is described by its intended use (own watchlist and history).

5.9 AI and Analytics Features

Analytics. The analytics in the application are the Media Types chart, the Viewing Progress chart and the summary tiles on the Dashboard. Zoho Analytics was not integrated.

AI features of the platform. No Zia or other AI feature of Zoho Creator is part of the delivered application.

AI assistance during the project. An AI assistant (Claude, by Anthropic) was used to help restructure this report from the project description and screenshots. All configuration was carried out in Zoho Creator by the student. (The student should edit this paragraph so that it states exactly what AI help was used.)

5.10 Pro-Code Extension

No pro-code extension (Zoho Catalyst, custom functions, JavaScript widgets or the Creator REST API) was used, because the required behaviour was covered by forms, workflows and short Deluge scripts. A future extension could use the Creator REST API to import titles from an external movie database, or use the Zoho Sheet connection to export the library.

5.11 Mobile and Portal Access

The builder offers desktop, tablet and phone previews. In the phone layout the Media_Library form, the Viewing_Activity form and the Dashboard open as single-column screens (Figure 5.19). The Deploy → Portal page lists the application CineTracks as a Public portal with Default authentication and 1 portal user, at the address https://rishikumarr2024csbsra282.zohocreatorportal.in (Figure 5.20). The Zoho Creator mobile app was not set up or tested.

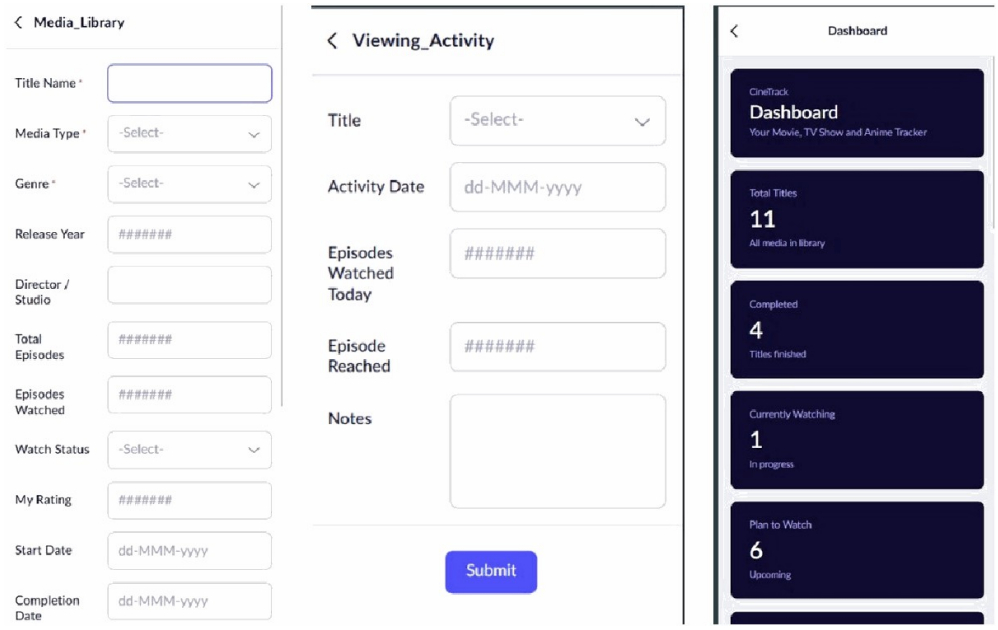


Figure 5.19: CineTrack in the Phone Layout – Media_Library, Viewing_Activity and Dashboard

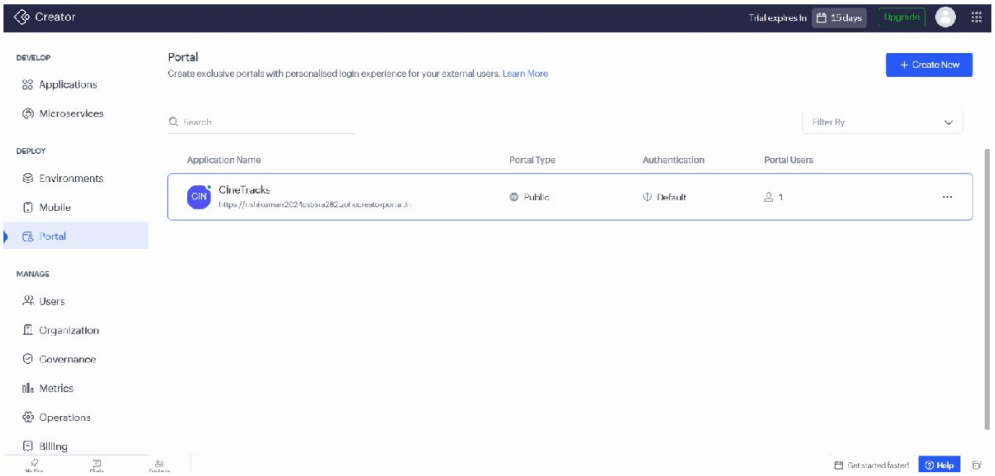


Figure 5.20: Portal Page Showing the CineTracks Portal (Public, Default Authentication, 1 User)

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was done on the live application while it was being built. Unit testing checked each form with sample titles and each report and chart by comparing it with the records. Integration testing checked the workflows by submitting titles and observing the stored status, the approval task and the e-mails received. Role-based testing by signing in as each role is not evidenced and is listed as not run. User acceptance testing with other users has not been recorded (Section 6.5).

6.2 Test Cases

Table 6.1: Test Cases

TC	FR	Description	Input / Steps	Expected	Actual	Status
TC-01	FR-01	Add a title with mandatory fields	Dorothy, Movie, Drama, 2026, Karthik Subbaraj	Record saved	Saved as Title ID 12 (Fig. 5.10, 5.12)	Pass
TC-02	FR-01	Mandatory fields left empty	Submit with Title Name empty	Form asks for value	Not executed	Not run
TC-03	FR-04	Default approval status	Submit a new title	Approval Pending	Approval task created (Fig. 5.10)	Pass
TC-04	FR-05	Auto complete	Episodes Watched = Total Episodes (e.g. Attack On Titan 94/94)	Watch Status = Completed	Completed (Fig. 5.5, 5.6)	Pass
TC-05	FR-06	Rating validation	My Rating = 11	Alert and cancel submit	Not executed	Not run
TC-06	FR-06	Valid rating saved	My Rating 9	Saved	Ratings 9 and 10 stored (Fig. 5.5)	Pass
TC-07	FR-07	Completed e-mail	Mark a title Completed	E-mail received	E-mails for Dorothy and Past Lives (Fig. 8.4)	Pass
TC-08	FR-08	Approval e-mail	Submit Dorothy	E-mail to approver	Received (Fig. 8.3)	Pass
TC-09	FR-08	Approve / Reject	Open Approval Summary	Task moves to Completed	Tasks 9, 10, 11 in Completed Tasks (Fig. 5.11)	Pass
TC-10	FR-09	Kanban board	Open Watch Progress Board	Cards by status	Shown (Fig. 5.6)	Pass
TC-11	FR-09	Dashboard	Open Dashboard	Tiles and charts	Shown (Fig. 5.7)	Pass
TC-12	FR-10	Permission sets and roles saved	Create sets and roles	Listed	Listed (Fig. 5.17, 5.18)	Pass
TC-13	FR-10	User role cannot edit	Sign in as User and try to edit	Edit not offered	Not executed	Not run
TC-14	FR-11	Scheduled reminder	Planned date reached	Reminder e-mail	Schedule shown Enabled (Fig. 5.14); reminder e-mail not captured	Not verified

6.3 Debugging Deluge Scripts

Debugging relied on the error or alert message shown by a form on submission, the behaviour of the live records after a test submission (for Auto_Complete_Title, comparing the status typed with the status stored), and the e-mails actually received. A script that saves in the workflow editor without syntax errors can still fail at run time, so each rule was checked with real data.

Test observations to record. Edit the following to match what you actually met while building: (1) a script that did not trigger because it was attached to the wrong form event; (2) a null value in Total Episodes (Movies have no episode count), which is why the scripts test for null before comparing.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Movies have no Total Episodes, so a comparison with a null value is unsafe.	Number field left empty.	Auto_Complete_Title checks for null and a value above 0 before comparing.	Closed
BUG-02	Reminder e-mail of Upcoming_Watch_Reminder not captured in testing.	The schedule is enabled but no e-mail was screenshotted.	Capture the e-mail on a planned date.	Open
BUG-03	Zoho Sheet connection has no references.	Connection configured but not used by any script.	None yet; listed as future work.	Open
BUG-04	Role-based sign-in tests not run.	Not evidenced.	To be run at the demonstration.	Open

The student should add any other defects actually met during development to Table 6.2.

6.5 User Acceptance Testing

No feedback from other users has been recorded. Table 6.3 is left for the student to complete after at least two people have tried the application.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Rishi Kumar R (developer)	Admin / Editor	Submitted Dorothy; the approval task appeared in Pending Tasks and the approval and completion e-mails were received.	None needed.
Rishi Kumar R (developer)	Editor	Attack On Titan (94/94) was set to Completed automatically and ratings 9 and 10 were saved. Movies have no episode count.	Null check added (BUG-01).
Rishi Kumar R (developer)	Admin	Dashboard, Kanban board and phone layout opened correctly. Reminder e-mail and role sign-in were not tested.	Left open (BUG-02, BUG-04).

Table 6.3 records the developer's own testing only; feedback from other users has not been collected.

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator provides a multi-stage deployment space (Deploy → Environments) with Stage and Production environments, and development takes place in the builder. The Environments page shows the application CineTracks with the message "No changes available", version 1.0 in Stage and version 1.0 in Production, both dated Nov 01, 2026, and the Publish menu offers Publish to Stage and Publish to Production (Figure 7.1). The rollback plan is to keep the previous version in the Environments page and republish it if a later change breaks the application.

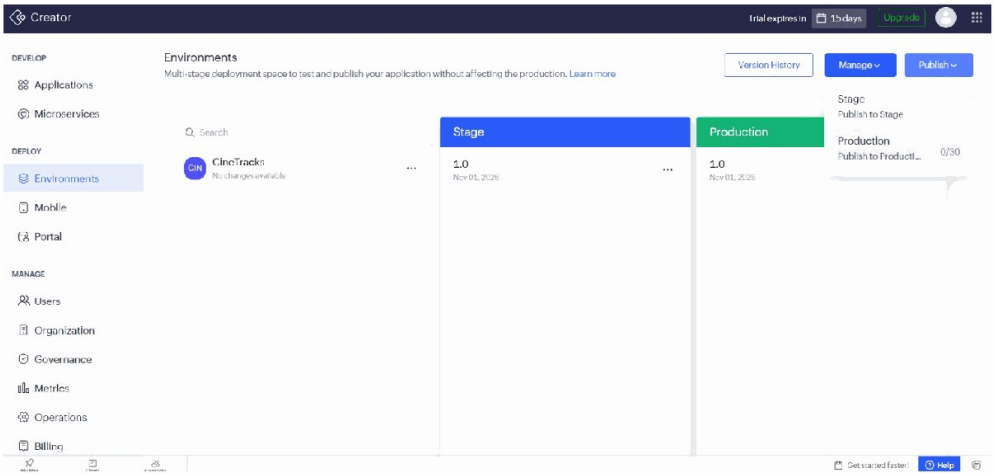


Figure 7.1: Environments Page Showing Version 1.0 in Stage and Production (01 Nov 2026)

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done
1	All test cases executed (TC-02, TC-05 and TC-13 not run; TC-14 not verified).	No
2	Sample/test data removed or isolated (sample titles remain).	No
3	Roles and permission sets defined (users assigned: to be confirmed).	Partial
4	Connections authorised (zoho_sheet_connection Connected).	Yes
5	Form workflows, schedule and approval process enabled.	Yes
6	Application published to Stage / Production and URL verified (Fig. 7.1).	Yes
7	Version / release notes recorded.	No

7.2 Performance Optimisation

No performance measurements were taken, so no improvement is claimed. Design choices

that help performance are that the dashboard charts use aggregate counts, the reports show a limited set of columns, and the scripts use no fetch statements or loops.

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Permission sets	Admin, CEO, Editor, User, Read, Write	Unauthorised changes to records
Role hierarchy	CEO → Admin → Editor, User (Settings → Roles)	Uncontrolled visibility between levels
Approval process	Watchlist_Title_Approval	Unreviewed titles entering the approved library
Mandatory fields and rating validation	Media_Library form and Validate_Media_Rating	Incomplete or out-of-range data
Connection-level access	zoho_sheet_connection: access turned on per application	Use of the connection by unrelated applications

Not applied: field-level encryption, IP or domain restrictions, two-factor authentication and record-level criteria. Authentication is by the platform's account sign-in.

7.4 Governance and Administration

Administration is done in Settings (Users, Permissions, Roles, Data Sharing). The Operations → Audit Trail page of Zoho Creator shows the change audit for CineTrack → Media_Library, listing the record additions and edits made by the application owner with their record IDs and times (Figure 7.2). It is the evidence for NFR-06. Naming follows a simple convention: one form per entity, reports named after their purpose and workflows named after what they do.

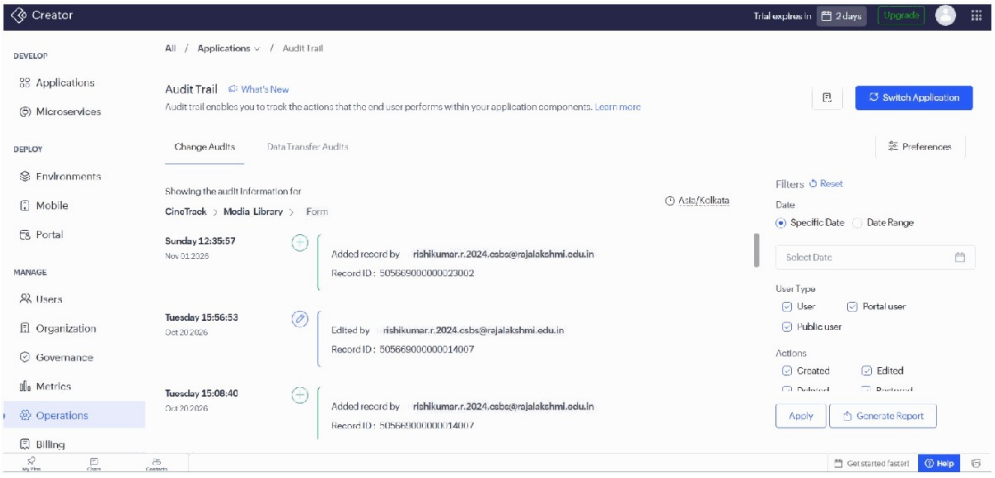


Figure 7.2: Audit Trail of the Media_Library Form

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The working application is presented through the Dashboard, the e-mail notifications sent by the workflows and the Zoho Sheet connection, followed by the approval scenario of the new title Dorothy. Each output is shown as a screenshot taken from the running application and explained below.

8.1.1 Dashboard

Figure 8.1 shows the Dashboard page built in the UI Builder. A row of five summary tiles at the top gives Total Titles, Completed, Currently Watching, Plan to Watch and Average Rating. In this screenshot the tiles read 10 total titles, 3 completed, 1 currently watching, 6 planned and an average rating of 9 out of 10 (Table 8.3). Below the tiles, the Media Types donut chart splits the library into movies, TV shows and anime, and the Viewing Progress bar chart counts titles under Plan To Watch, Watching and Completed. The embedded Viewing_Activity Report lists the recorded viewing sessions on the same page, and the Navigate panel gives quick actions for My Library, Add Title, Viewing Activity and Watchlist.

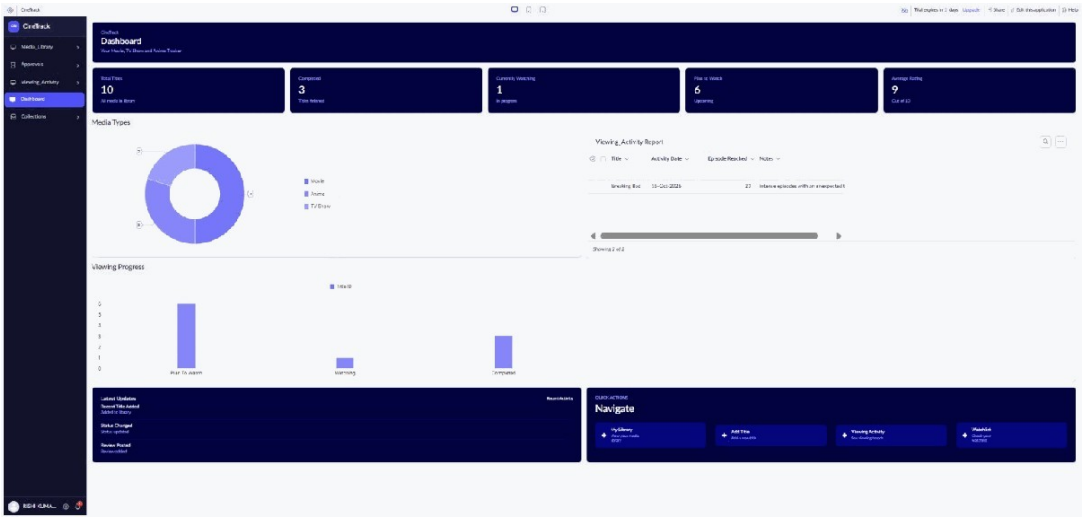


Figure 8.1: CineTrack Dashboard – Summary Tiles and Charts

The values on the tiles are calculated from the Media_Library records, so they change when a title is added or its Watch Status changes. The viewer therefore does not have to count titles by hand, which was one of the pain points in Section 1.2.

8.1.2 Dashboard in the Phone Layout

Figure 8.2 shows the same Dashboard opened in the phone layout. The tiles are stacked in a single column and read 11 total titles, 4 completed, 1 currently watching and 6 planned. These figures are higher than those of Figure 8.1 because more titles were added and marked Completed after the first screenshot, which shows that the dashboard follows the stored data and is not a fixed picture. The Zoho Creator mobile app itself was not set up or tested (Section 5.11); this is the builder's phone preview.

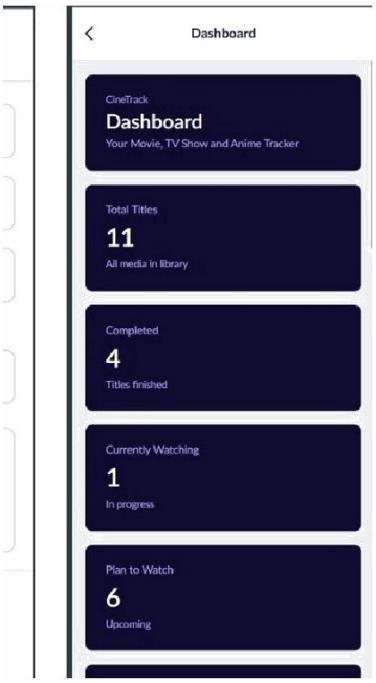


Figure 8.2: Dashboard in the Phone Layout (11 Titles, 4 Completed)

8.1.3 E-mail Notifications

Two kinds of e-mail were received in the inbox, both sent by the platform from notifications@trial.zohocreatormail.com.

Approval request e-mail. When Dorothy was submitted, the approver received the e-mail shown in Figure 8.3 with the subject "New Title Submitted for Approval: Dorothy". It is addressed to "Dear Approver" and lists the Title (Dorothy), Genre (Drama), Media Type (Movie), Release Year (2026) and Status (Pending Approval), and asks the approver to review and approve or reject the submission. It is signed "CineTrack System". The same title also appears as Title ID 12 under Approvals → Pending Tasks (Figure 5.10), where the approver opens the Approval Summary and chooses Approve or Reject (Figure 5.12).

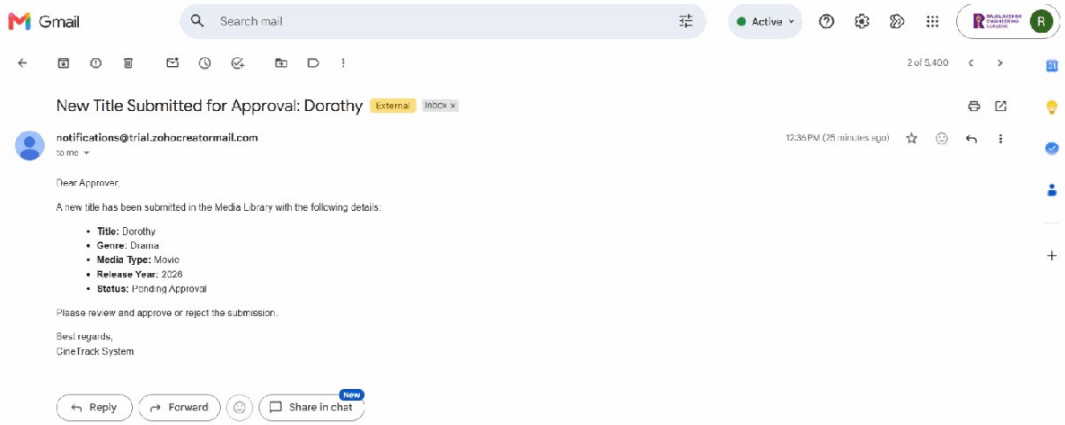


Figure 8.3: Approval Request E-mail – "New Title Submitted for Approval: Dorothy"

Completion e-mail. When a title is marked Completed, the user receives an e-mail with the subject “CineTrack: Title Completed” (Figure 8.4). The first message states that Dorothy was marked as Completed in CineTrack and that the viewing library has been updated; the second states that Past Lives was marked as Completed. Two e-mails in the same thread show that the notification is sent for each title and not only once.

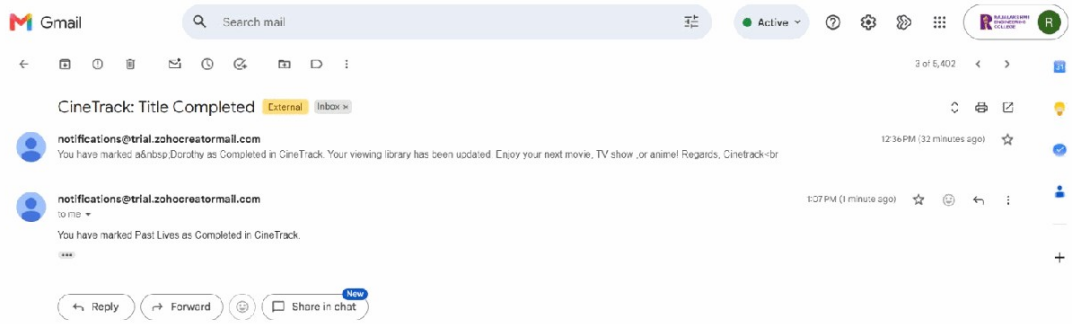


Figure 8.4: Completion E-mails – “CineTrack: Title Completed”

These e-mails are the visible result of the sendmail actions in the form workflows (Table 5.2). The reminder e-mail of the schedule Upcoming_Watch_Reminder is not shown because it was not captured (Section 5.6).

8.1.4 Zoho Sheet Connection

Figure 8.5 shows Microservices → Connections in Zoho Creator, where Zoho Sheet Connection (link name zoho_sheet_connection) is listed with the status Connected. Figure 8.6 shows its details: the service is Zoho Sheet, the permissions include All Data Operations and Create Record, the authorization is the same for all environments, and the authorization owner is the Administrator. Under Connection Access the switch for CineTrack is turned ON, which allows CineTrack to use the connection.

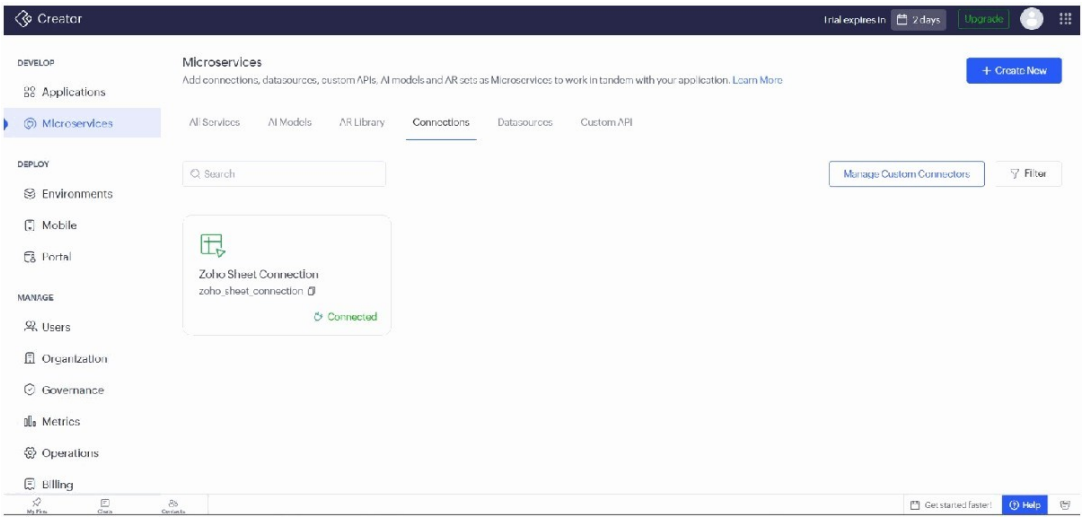


Figure 8.5: Zoho Sheet Connection Listed as Connected

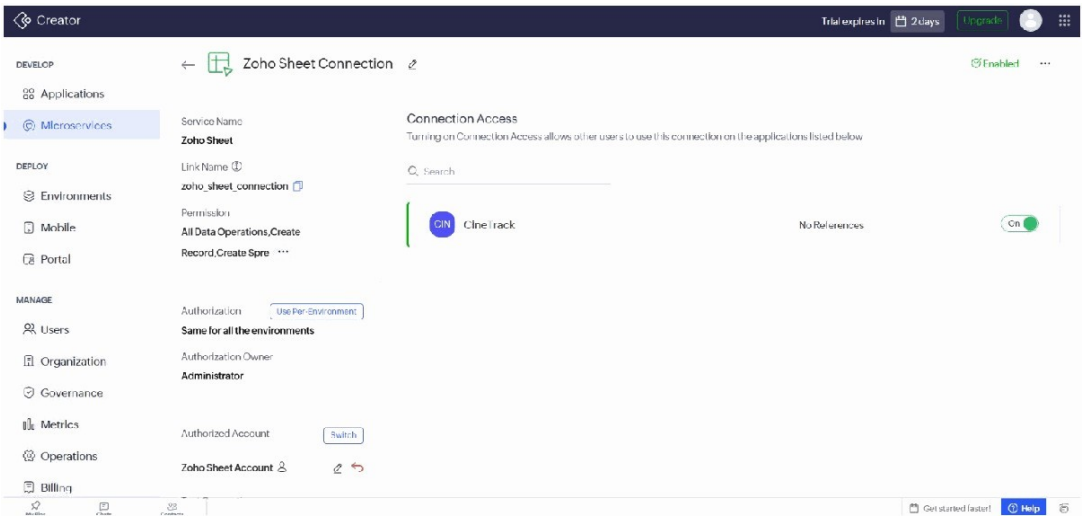


Figure 8.6: Zoho Sheet Connection Access Enabled for CineTrack

The same page shows “No References” beside CineTrack. This means that no Deluge script, workflow or report uses the connection yet, so the integration is configured and authorised but not exercised. It is therefore reported as Partial in Appendix A. A planned use is a script that writes the library to a Zoho Sheet spreadsheet as a backup or export (Section 9.2); that script has not been written and no export was tested.

8.1.5 Summary of Output Figures

Taken together, the figures of this section support the following requirements. Figures 8.1 and 8.2 show the dashboard and its phone layout (FR-09 and the objective of analysing viewing habits). Figure 8.3 shows the approval request e-mail (FR-08). Figure 8.4 shows the completion e-mails (FR-07). Figures 8.5 and 8.6 show that the Zoho Sheet connection is authorised and available to CineTrack, although no script uses it yet (Appendix A, Partial).

8.1.6 Scenario: Submitting the Title Dorothy

The approval scenario runs as follows. In step 1 the title is submitted in the Media_Library form. In step 2 an approval task appears under Approvals → Pending Tasks with Title ID 12, Title Name Dorothy and Media Type Movie (Figure 5.10), and the approver can open the Approval Summary and choose Approve or Reject (Figure 5.12). In step 3 the approver receives the e-mail of Figure 8.3. In step 4 the completion e-mail of Figure 8.4 is received when a title is marked Completed. In step 5 the Dashboard (Figures 8.1 and 8.2) shows the updated counts.

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Centralised library of movies, TV shows and anime	Media_Library form and reports (5.2, 5.3)	Achieved (Fig. 5.1, 5.4)
Track viewing progress	Episodes fields, Viewing_Activity, Kanban board	Achieved (Fig. 5.2, 5.6)
Watchlists and collections	Watchlist Collections form	Form built (Fig. 5.3)
Rate and review	My Rating, My Review, rating validation	Achieved (Fig. 5.9)
Automate tasks with Deluge	Auto_Complete_Title, Validate_Media_Rating, approvals	Achieved (Fig. 5.8, 5.9, 5.10–5.12)
E-mail notifications and reminders	Approval and completion e-mails; scheduled reminder	E-mails achieved (Fig. 8.3, 8.4); schedule enabled (Fig. 5.14), reminder e-mail not captured
Analyse viewing habits	Reports and Dashboard	Achieved (Fig. 5.4–5.7)

8.3 Comparison with the Existing System

No timing or error-rate measurements were made, so Table 8.2 compares the existing practice and the application qualitatively and no percentage improvements are claimed.

Table 8.2: Existing System versus CineTrack

Parameter	Existing System	CineTrack
Recording a title	Notes or memory; structure varies	One form with fixed fields and drop-downs
Progress tracking	Not kept in one place	Episodes watched and viewing activity per title
Ratings	Unchecked or not recorded	Validated between 1 and 10, with a review box
Completion	Remembered by the viewer	Automatically set to Completed and e-mailed
Overview	Scroll through several apps	Dashboard, Kanban board and charts on one page

Table 8.3 records the quantities that can be read from the dashboard screenshot (Figure 5.7). They describe the sample data and are observations, not performance measurements.

Table 8.3: Observed Application State from the Screenshots

Measure	Observed Value	Source
Total titles	10	Dashboard tile
Completed	3	Dashboard tile
Currently watching	1	Dashboard tile
Plan to watch	6	Dashboard tile

Measure	Observed Value	Source
Average rating	9 out of 10	Dashboard tile
Records in Media_Library Report	9 of 9	Fig. 5.4

8.4 Limitations of the Low-Code Approach

- Editions and trial limits. The application was built on a trial edition that expires soon, so some features may not be available after it ends.
- Platform dependency. The application exists only inside Zoho Creator and the logic is written in Deluge, which is specific to Zoho.
- Lookup display. A lookup shows only the field chosen as display value, so a clear title name field is needed for readable lookups.
- Scheduling and integration. The reminder e-mail of the schedule still has to be verified, and the Zoho Sheet connection is not yet used.
- Scripting scope. Deluge covered every need of this project, but its statement and API- call limits were not tested because the scripts are small.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application is no-code configuration: the forms, lookups, reports, the Kanban board, the dashboard, the roles and permission sets and the approval process. Low-code was needed for the Deluge scripts for completion, rating validation and notifications. Pro-code was not needed. It would be justified if CineTrack had to import titles from an external movie database through an API, or to recommend titles using heavier processing.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

CineTrack transforms the way users organise and track their entertainment content. By combining a media library, viewing history, personal collections, reports and automated workflows, the application provides an organised and efficient media tracking experience. It also demonstrates the practical implementation of low-code application development using Zoho Creator, with data management, reports, automation, an approval process and role-based access.

The work also has limits that are recorded openly: the reminder e-mail of the schedule is not yet captured, the Zoho Sheet connection is not yet used, role-based sign-in testing and user feedback are not recorded, and no timing measurements were made. These items are the first to address.

9.2 Future Enhancements

- Capture and verify the reminder e-mail of Upcoming_Watch_Reminder on a planned date.
- Use the Zoho Sheet connection to export the library or import titles.
- Assign real users to the roles, add record-level criteria so that a User sees only their own watchlist, and test every role.
- Add a validation that Completion Date is not before Start Date.
- Add a customer portal or mobile configuration for quick updates from a phone.
- Import title details from an external movie database through the Creator REST API.

9.3 Learning Outcomes and Career Relevance

Building this application gave practice in data modelling with forms and lookups, validations, list, Kanban and chart reports, UI Builder pages, form workflows, Deluge scripting, approvals, connections, roles and permission sets, and the audit trail. It also showed how testing and debugging work in a low-code platform. These skills relate to the roles of citizen developer, low-code developer, solutions consultant and automation engineer.

REFERENCES

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[2] M. A. Al Alamin, S. Malakar, G. Uddin, S. Afroz, T. B. Haider and A. Iqbal, “An Empirical Study of Developer Discussions on Low-Code Software Development Challenges,” arXiv:2103.11429, 2021. [Online]. Available: https://arxiv.org/abs/2103.11429

[3] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/

[4] Zoho Corporation, “Deluge Script Reference.” [Online]. Available: https://www.zoho.com/deluge/

Note: only sources actually cited in the text are listed.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Y = implemented, Partial = implemented only in part, N = not implemented. Re-check against the live application before the demonstration.

Table A.1: Feature Coverage Checklist

Unit	Feature	Done	Where / Note
I	Application created and menu organised	Y	Sections Media_Library, Approvals, Viewing_Activity, Dashboard, Collections
II	Forms with ≥ 8 field types	Y	Single Line, Drop Down, Number, Date, Multi Line, Lookup, etc.
II	Lookup field	Y	Title in Viewing_Activity and Watchlist Collections
II	Subform	N	Not used
II	Field validations	Y	Mandatory fields; rating validation script
II	List report	Y	Media_Library Report, Media Watchlist
II	Kanban report	Y	Watch Progress Board
II	Calendar / Timeline / Map report	N	Not used
II	Pivot / chart report	Y	Viewing Statistics and dashboard charts
II	Page built with UI Builder	Y	Dashboard
II	Deluge – form events	Y	Validations on form submission
II	Email notifications	Y	Approval and completion e-mails
II	Form workflow	Y	Five automations
II	Scheduled workflow	Y	Upcoming_Watch_Reminder (Fig. 5.14)
II	Approval process	Y	Watchlist_Title_Approval
II	Blueprint / Batch workflow	N	Not used
II	Testing and debugging evidence	Y	Chapter 6
III	Integration with a Zoho service	Partial	Zoho Sheet connection configured, not referenced
III	REST API / invokeurl / webhook	N	Not used
III	Users, roles and permission sets	Y	Chapter 5.8
III	Record-level criteria / portal	Y	Portal listed (Fig. 5.20)
III	Zia / AI feature	N	Not used
III	Analytics	Y	Dashboard charts
IV	Deployment across environments	To confirm	Add from Environments page
IV	Governance (audit trail)	Y	Audit Trail screenshot
V	Pro-code extension	N	Not used; justified in 5.10
—	Mobile / portal	Partial	Phone layout (Fig. 5.19); portal (Fig. 5.20)

APPENDIX B – COURSE OUTCOME MAPPING

The self-assessed level (1 – Low, 2 – Medium, 3 – High) is left for the student; the examiner may revise it.

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self	Examiner
CO1	Explain the principles, architecture and business value of low-code development using Zoho Creator.	Ch. 1, 2		
CO2	Develop business applications using Zoho Creator forms, reports, workflows and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6		
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections and role-based security.	Ch. 4.6, 5.7–5.9		
CO4	Deploy, secure and manage Zoho Creator applications across environments following governance best practices.	Ch. 7		
CO5	Evaluate low-code strategies, AI-assisted development and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3		

APPENDIX C – DELUGE SOURCE CODE

The two scripts below are as visible in the workflow editor screenshots. Paste the other three scripts from the application where indicated.

C.1 Validations on form submission – Auto_Complete_Title (Media_Library)

```
if(input.Total_Episodes != null)
{
    if(input.Total_Episodes > 0 && input.Episodes_Watched != null)
    {
        if(input.Episodes_Watched >= input.Total_Episodes)
        {
            input.Watch_Status = "Completed";
        }
    }
}
```

C.2 Validations on form submission – Validate_Media_Rating (Media_Library)

```
if(input.My_Rating != null)
{
    if(input.My_Rating < 1 || input.My_Rating > 10)
    {
        alert "Rating must be between 1 and 10.";
        cancel submit;
    }
}
```

C.3 Default approval status (Media_Library)

```
// Paste the script saved in the application
```

C.4 Completed title notification (Media_Library)

```
// Paste the script saved in the application
```

C.5 Watchlist approval request e-mail (Media_Library)

```
// Paste the script saved in the application
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Sign in to Zoho Creator (India data centre, creator.zoho.in) with your Zoho account and open CineTrack: https://rishikumarr2024csbsra282.zohocreatorportal.in. The menu shows Media_Library, Approvals, Viewing_Activity, Dashboard and Collections. The application can also be opened in the phone layout (Figure 5.19).

D.2 Instructions for the Admin

1. Open Approvals → Pending Tasks and open a task, or follow the link in the e-mail "New Title Submitted for Approval".
2. Check the Approval Summary and click Approve or Reject.
3. Open Dashboard to see the totals, charts and recent activity.
4. Manage roles and permission sets in Settings → Users.

D.3 Instructions for the Editor

1. Open Media_Library → Media_Library, enter Title Name, Media Type and Genre (mandatory) and any optional details, and click Submit.
2. Keep My Rating between 1 and 10; otherwise the form will not submit.
3. Open Viewing_Activity, choose the title, enter the date and episodes watched, and submit.
4. Open Collections to add a title to a watchlist collection.
5. Use Media_Library → Watch Progress Board to see titles by status.

D.4 Instructions for the User

1. Open Media_Library → Media Watchlist to see titles, status and ratings.
2. Open Viewing_Activity to see the viewing history.
3. Open Dashboard for the overall picture.

The Editor and User instructions describe the roles as designed; confirm them against the live application.